

1           Physical Primitives of Mind: Modeling  
2   Consciousness via Symmetries and Landauer  
3           Erasure

4                   Sebastian Bobadilla-Suarez  
                  `sebastian.bobadilla.s@gmail.com`

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6                   **Abstract**

7           We present a mathematical synthesis and formal Lean 4 verifica-  
8   tion of the dependencies in a candidate route from established phys-  
9   ical primitives to cognitive attractor states, without introducing new  
10   fundamental physics. Starting from the finiteness of a localized sys-  
11   tem’s phase space, we construct a framework for predictive processing;  
12   the spacetime symmetries are the setting rather than a premise, and  
13   no theorem consumes them. Spontaneous symmetry breaking gener-  
14   ates necessary topological boundaries, and finite phase space sub-  
15   jects these boundaries to Landauer’s thermodynamic erasure. Agency

16 arises when systems evolve along paths of least action, minimizing en-  
 17 tropy production via internal structural resonance with external per-  
 18 turbations. Scaling this dynamic to macroscopic levels aligns with  
 19 emerging paradigms in neuroscience—specifically classical electrody-  
 20 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-  
 21 tion, culminating in a topological fixed point: a unified “Self.” We  
 22 state the claim narrowly, because the subject matter invites a wider  
 23 reading than the work supports: no new physics is derived here. Every  
 24 physical result the chain invokes is established—Picard–Lindelöf, Kib-  
 25 ble’s classification of topological defects, Landauer’s erasure bound,  
 26 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-  
 27 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed  
 28 point. What is new is that they are stated in one formal language,  
 29 their consistency mechanically checked, and the points where the chain  
 30 relies on stipulation rather than derivation located exactly. One condi-  
 31 tional theorem composes eight named hypotheses to its endpoint; the  
 32 eight unproved steps are its arguments rather than sentences about  
 33 it, and they are jointly satisfiable rather than jointly empty. The de-  
 34 velopment declares no axioms and every structure carrying physical  
 35 content is inhabited by a witness; the synchronization step is proved  
 36 from a dynamics for initial data inside an explicit basin and assumed  
 37 outside it, and the identification of a global section of the probability  
 38 sheaf with the unity of experience is a stipulation, marked as such  
 39 throughout rather than absorbed into the derivation. After the scalar

coarse-graining theorem, the framework posits a continuum kernel and identifies it with the brain’s endogenous electromagnetic field; these are respectively a modelling assumption and an empirical commitment, and four conditions would falsify the latter. The framework is nonetheless falsifiable in one place that is its own and no rival’s: recovery from sudden awakening should follow a delayed sigmoid rather than an exponential, because the variable gating it is astrocytic volume on a timescale of seconds to minutes rather than cortical phase realignment on a timescale of milliseconds; and, more sharply, the concentration and the coherence of the phase distribution must collapse onto a single parameter-free curve, the Bessel ratio, throughout that recovery. As a corollary, we show why rigid von Neumann architectures (e.g., GPUs) are thermodynamically disadvantaged or structurally constrained from reaching these specific continuous thermodynamic attractors.

## 1 Introduction

The pursuit of a physical theory of consciousness typically involves searching for the biological correlates of mind or treating subjective experience as an abstract computational algorithm. This paper asks a more fundamental question: If we assume only basic physical primitives—continuous symmetries (the Poincaré group), geometry, and thermodynamics—is the emergence of a unified, perceiving entity a natural physical attractor?

62 We propose a framework in which the structural properties of conscious-  
 63 ness are organized as candidate physical attractor states. Its narrative runs  
 64 from continuous symmetries and boundaries, through dissipation and phys-  
 65 ical resonance, to the macroscopic electromagnetic unity proposed by field  
 66 theories of mind. This is not a derivation from basic principles: the for-  
 67 mal result is a conditional composition schema whose eight unproved arrows  
 68 are explicit assumptions. To avoid critiques of “math theatre,” we delin-  
 69 eate between literal physical mechanisms (e.g., thermodynamic erasure via  
 70 Landauer’s principle, boundaries from spontaneous symmetry breaking) and  
 71 theoretical mappings, utilizing Lean 4 formalization as our mathematical  
 72 anchor. We adopt a perspective aligned with Russellian Monism: objec-  
 73 tive measurements of macroscopic electromagnetic fields constitute the third-  
 74 person “outside,” while the system’s internal physical deformation to reach  
 75 structural resonance constitutes the first-person intrinsic experience.

## 76 **What is new here, and what is not**

77 A framework paper about consciousness invites a particular suspicion, and it  
 78 is better answered at the start than at the end. **No new physics is derived**  
 79 **here.** Every physical result the chain invokes is established, and most of  
 80 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics  
 81 (Coddington & Levinson, 1955); Kibble’s classification of topological defects  
 82 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound  
 83 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative

84 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a  
85 dissipating trajectory (Barbalat, 1959); the threshold  $K_c = 2D$  for the noisy  
86 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,  
87 Bell and Crooks relating dissipation to nonpredictive information (Still et  
88 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating  
89 dissipated work with the relative entropy between forward and time-reversed  
90 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,  
91 1922). A physicist reading this paper will recognize all of it and would  
92 establish the point in three paragraphs if we did not. Conceding it costs  
93 nothing.

94 What is new is of a different kind, and it is a smaller claim than a theory  
95 of consciousness usually makes. These results have been stated in one formal  
96 language, their mutual consistency has been mechanically checked, and the  
97 places where the chain relies on stipulation rather than derivation have been  
98 located exactly. The value of a machine-checked chain is not that any link  
99 is novel; it is that the claim being made *about* the chain is itself checkable,  
100 and we state it in the form a reader can check rather than in the form that  
101 asks to be believed. Four things. The development declares **no axioms**:  
102 **#print axioms** on any result reports only **propext**, **Classical.choice** and  
103 **Quot.sound**. Every physical postulate is a field of a class that each model  
104 discharges for itself rather than a global assertion, and every such class is  
105 inhabited by an explicit witness, so no theorem is vacuous for want of a  
106 model. The links form a single *conditional composition theorem* — **chain**,

107 in `Chain.lean` — with **eight named hypotheses**: four formalization gaps,  
 108 two modelling assumptions and two physical commitments. Its conclusion is  
 109 `UnifiedSelf`: the particular section produced by sheaf gluing is the unique  
 110 fixed point of self-prediction, rather than `Unity` and `Self` being witnessed  
 111 by unrelated sections. The  $n8 \rightarrow n9$  hypothesis explicitly carries both the  
 112 nonlinear Lipschitz law and this state identification; neither is derived from  
 113 field dynamics. The theorem `chain_hypotheses_jointly_satisfiable` dis-  
 114 charges all eight on one toy substrate, which rules out inconsistency but is not  
 115 evidence that one cortical mechanism realizes the composition. The seams  
 116 — where physics stops and interpretation begins — are thereby fixed in the  
 117 source rather than negotiable in the prose.

118 The schema exposes two missing arrows. The capacity bound of the first  
 119 link has no hypotheses at all, so an implication from it to the existence of a  
 120 boundary is equivalent to assuming the boundary outright: those two links  
 121 are independent starting points and not consecutive steps. What finiteness  
 122 does imply directly is the heat cost, skipping the boundary entirely. Neither  
 123 of the two arrows that are theorems is one the framework’s usual narration  
 124 draws. Section 2.1 reports the corresponding soundness result: three tempt-  
 125 ing global axiom shapes each prove **False**, from one shared root cause.

126 The framework is not thereby unfalsifiable, and one of its predictions is  
 127 its own rather than a rival’s. Section 11 argues that recovery from sudden  
 128 awakening should follow a *delayed sigmoid* rather than an exponential, be-  
 129 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid

clearance, on a timescale of seconds to minutes, while the cortical rhythms whose realignment is usually taken to constitute the recovery move in milliseconds. That is a claim about functional form; it is fittable against data of a kind already collected; and a purely electrical account of the same recovery predicts a different form. Because a delayed sigmoid is what any slow saturating gate would give, the section also states the sharper claim the same mathematics supports: at every instant of the recovery the concentration and the coherence of the phase distribution must lie on the fixed curve  $r = I_1(a)/I_0(a)$ , the same curve for every subject, with no free parameters and with the threshold read off the ratio  $a/r$  rather than fitted.

## 2 Formalization in Lean 4

A core contribution of this work is the machine-checked dependency accounting of the framework’s conditional composition schema using the Lean 4 interactive theorem prover. This approach builds on recent efforts to formalize consciousness and brain semantics, such as the ‘OnlyOne’ system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué, 2026). It is necessary to explicitly distinguish which claims have been mathematically derived as theorems within the proof assistant, and which remain irreducible physical postulates. Figure 1 shows the schema with the status of every node *and of every arrow* marked, and Table 1 states in seven rows what each link assumes. The complete claim-by-claim map to Lean identifiers, which runs to

several pages, is Table S1 in the supplement; the main text does not depend on reading it.

## 2.1 Soundness of the formalization

Formalization is only as trustworthy as the axioms it rests on, and an axiom that is inconsistent silently makes every theorem downstream of it vacuous while leaving the build green. We report a case of exactly this, found by auditing our own development, because the failure mode is easy to reproduce and we have not seen it discussed in the applied-formalization literature.

The failure has one shape, and naming it is more useful than cataloguing its instances: *an axiom that constrains a symbol it does not itself bind*. Such a declaration is not a postulate about one system. It is a constraint on every model of the class the symbol belongs to, including models built to violate it, and a class that has such a model is thereby inconsistent. Three concrete cases, each of which proves `False` on its own:

- `spontaneous_symmetry_breaking_pointwise_min` took the potential-energy functional as an *implicit* argument occurring only in its hypothesis `PotentialEnergy phi = V v0`. Instantiating that functional with the constant map  $\lambda_. V(v_0)$  discharges the hypothesis by `rfl` and yields the conclusion for an arbitrary field. With  $V(v) = v^2$ ,  $v_0 = 0$  and  $\phi \equiv 1$  this proves  $1 = 0$ .
- `landauer_heat_eq` equated the free class field `heat_dissipation` with

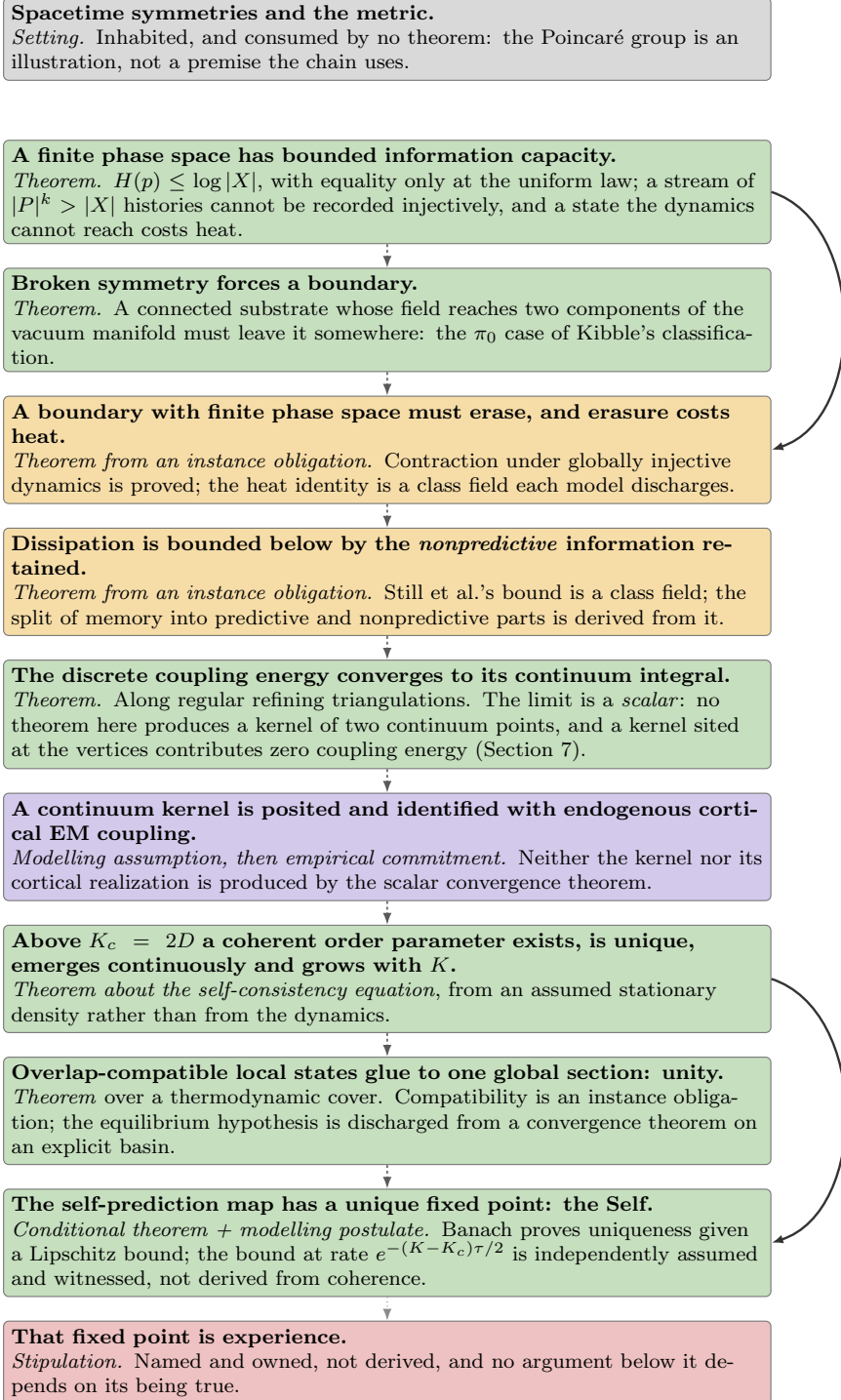


Figure 1: The conditional composition<sup>9</sup> schema, with eight assumed vertical arrows and the status of each node and of each arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of chain (**Chain.lean**): an implication between the two node statements which the development does not prove

| Claim                                                                                                                       | Status                                            | What is assumed                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A finite phase space has bounded information capacity                                                                       | Theorem                                           | Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log  X $ , with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.                                                                                                          |
| Broken symmetry forces a boundary                                                                                           | Theorem                                           | Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.                                                                                                                                                                                                                                                                               |
| Erasure costs heat                                                                                                          | Theorem + class field                             | Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.                                                                                                                                                                                                                                          |
| Dissipation is bounded below by nonpredictive information                                                                   | Theorem + class field                             | Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.                                                                                                                                                                                                                                                                     |
| The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field | Theorem; <b>then empirical</b>                    | Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7). |
| Above $K_c = 2D$ a unique coherent state exists, and overlap-compatible patches glue into a unique global section           | Theorem<br><br><br><br><br><br><br><br><br><br>10 | Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum whose synchronized patches agree on overlaps; the first is derived from a convergent trajectory, the second is an instance obligation throughout.                                                |
| The coherent order parameter carries a unique reflexive fixed point:                                                        | Conditional theorem + modelling                   | The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate                                                                                                                                                                                                                                                                                                |

172 a bath-entropy difference *for every instance* of the class. An instance  
 173 with zero heat dissipation and a bath that gains one bit gives  $0 = \log 2$ .

174 • `kl_bound_axiom` asserted  $\sigma \geq D_{KL}(P\|Q)/\Delta t$  for *all*  $P$  and  $Q$ , while  $\sigma$   
 175 is a single real fixed by a class field. Since  $D_{KL}$  is unbounded above, a  
 176 point mass against a uniform distribution refutes it.

177 Two further declarations of the same shape are worth recording be-  
 178 cause their refutation is different in kind. `phase_invariant_periodic` and  
 179 `sync_to_section_eq`, as global assertions pinning free fields of `LocalSectionSynchronization`,  
 180 are refutable as soon as the probability presheaf admits two distinct global  
 181 sections — so the refutation needs no arithmetic at all, only that the sheaf is  
 182 not trivial. Neither is a postulate in the development: the periodicity condi-  
 183 tion is an obligation of the constructor that builds a cover from an invariant  
 184 measure, and `sync_to_section_eq` is a theorem about the global section the  
 185 sheaf condition produces (Derivation 5).

186 The rule that prevents it is uniform, and we state it as the rule we follow:  
 187 *a physical postulate that mentions a class field must be a field of that class,*  
 188 *never a standalone axiom quantified over all instances.* A postulate is then  
 189 an obligation each model discharges, not a global claim about every model.  
 190 Under that rule the development declares no axioms at all. Two of the  
 191 statements above are not postulates in the first place — pointwise vacuum  
 192 minimization is proved from  $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$  a.e., and Gibbs’  
 193 inequality is Mathlib’s — and the genuinely physical ones, Landauer’s heat

194 identity and the two information bounds, are instance obligations.  
 195 This makes non-vacuity the live question, since an uninhabitable class  
 196 makes its theorems vacuous just as surely as an inconsistent axiom did.  
 197 `Examples.lean` therefore constructs witnesses: a one-bit erasure system with  
 198 a genuine bath that discharges `heat_eq` and reproduces  $\Delta Q = k_B T \ln 2$ ; a  
 199 perfectly resonant system discharging `kl_bound`; and a scalar field on a one-  
 200 point spacetime discharging `ActionPrinciples`. Derivation 5's gluing struc-  
 201 tures are the most demanding case and are witnessed too: `Examples.lean`  
 202 builds a `ThermodynamicCover` over a three-site substrate carrying the dis-  
 203 crete topology and its Borel  $\sigma$ -algebra, covered by two *overlapping* open  
 204 patches whose intersection is a single shared site. The local sections are  
 205 germ families of a phase-indexed finite measure  $\mu_\theta = (1 + \cos \theta)^+ \delta_{\text{mid}}$  pushed  
 206 into the sheafification of the probability presheaf; the family is provably  
 207 non-constant in  $\theta$  (`phaseMeasure_not_const`), so the periodicity obligation  
 208 of the constructor `LocalSectionSynchronization.ofInvariantMeasure` is  
 209 discharged by an argument about `cos` rather than satisfied trivially, and  
 210 `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`  
 211 rather than assumed. `global_section_from_thermodynamics` then fires on  
 212 this instance and glues the two patch-local sections into a unique global one.  
 213 Two further covers answer the follow-up question of how non-degenerate such  
 214 a witness can be made. `Examples.lean` §13 has a negative half worth record-  
 215 ing: the phase field can be made non-constant as a function, but not as a  
 216 physical state, because `thermodynamic_equilibrium` forces every instance

217 to a phase-locked configuration. `Examples.lean` §14 is built the other way  
 218 round, from two patches with independently chosen mass profiles that meet  
 219 only on the site they share, and the section the gluing returns is neither of  
 220 them. That is the same kind of result as `contracting_implies_const` in  
 221 Derivation 6 — a proof that the witness is as strong as the class permits,  
 222 rather than a witness that could be improved by trying harder.

223 What the witnesses establish is that the classes are inhabitable, so the  
 224 theorems about them are not vacuous. What they do not establish is that  
 225 a cortex is such an instance. In the cover above the phase field is already  
 226 constant — every patch locked at  $\theta = 0$  — which is precisely what makes  
 227 the equilibrium obligation dischargeable; getting a physical system into that  
 228 configuration is the content of the informal dynamical argument, not of the  
 229 Lean development.

230 The same audit turned up a defect of a different kind, worth record-  
 231 ing because it is not an axiom problem and would not have been caught  
 232 by `#print axioms.mesh_refinement_convergence` — the claim that dis-  
 233 crete edge weights converge to the continuous energy as the mesh is refined  
 234 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was  
 235 not merely unproved but false. Nothing required the edge regions to cover  
 236 the manifold, and  $\text{diam } \emptyset = 0$ , so the triangulation whose edge regions are  
 237 all empty satisfied the mesh hypothesis while contributing nothing to the  
 238 sum, forcing  $|0 - \int f| < \varepsilon$  for every  $\varepsilon > 0$ . Its binder was dead as well: it  
 239 quantified over a triangulation and then resolved the vertex type by instance

240 search rather than through the bound variable, so the variable was never  
 241 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-  
 242 gence is now stated along a *sequence* of triangulations of a named region  
 243  $S$ , each required to be regular — edge regions measurable, without self-  
 244 loops, disjoint across distinct edges, covering  $S$ , and anchored to the sim-  
 245 plicial data in the sense that a vertex lies in the closure of every region  
 246 of an edge at it. Closure rather than membership is forced: cells are dis-  
 247 joint, so a shared vertex belongs to at most one of them. The manuscript’s  
 248 discrete energy  $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$ , which samples only the first endpoint  
 249 and is therefore not symmetric, is shown to equal exactly the midpoint rule  
 250  $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$  over unordered edges; without that identifica-  
 251 tion, reading the double sum as a Riemann sum would be a category error.  
 252 At fixed mesh the error is at most  $\varepsilon \mu(S)$  with  $\varepsilon$  a modulus of continuity of  
 253 the integrand, and convergence follows for uniformly continuous integrands  
 254 as the mesh shrinks. The regularity conditions pull against each other, so  
 255 `Examples.lean` §6 discharges them with the uniform partition of  $[0, 1)$  into  
 256 half-open cells and checks that the approximation genuinely moves —  $\frac{1}{2}$  at  
 257  $N = 1$  and  $\frac{1}{4}$  at  $N = 2$  for the tent function  $x \mapsto |x - \frac{1}{2}|$  — rather than  
 258 converging trivially. What is proved is convergence, not the  $O(1/N^2)$  rate  
 259 the accompanying simulation measures; that would need a midpoint error  
 260 term with a second derivative, which we have not developed.

261 A third defect was one of bookkeeping across files, and it is the kind that  
 262 survives any number of green builds because each half is individually cor-

263 rect. The development carried *two* Kuramoto potentials. Lyapunov descent  
 264 along the dynamics,  $\dot{V} = -\sum_i \dot{\theta}_i^2$ , was proved for the full potential  $V(\theta) =$   
 265  $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$ , while the characterization of the minimum  
 266 — phase-locked in both directions — and everything downstream of it, in-  
 267 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing  
 268 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`  
 269 now proves that as soon as one  $\omega_i \neq 0$  the full potential is unbounded below  
 270 — walk  $\theta$  out along  $\omega$  and the bounded cosine sum cannot compensate — so  
 271 it has no minimum for phase-locking to attain, and the sentence “the phase-  
 272 locked state minimizes the Lyapunov potential” was, read literally, false of  
 273 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`  
 274 closes this by formalizing the standard reduction  $\theta_i \mapsto \theta_i - \Omega t$ . For identical  
 275 natural frequencies  $\omega \equiv \Omega$ , a trajectory maps to a trajectory of the zero-  
 276 frequency system — which is exactly the system Derivation 5 already worked  
 277 with — the two potentials coincide there, and the descent result transports  
 278 onto the coupling potential whose minimizers are the phase-locked states.  
 279 Phase differences, phase-locking and the magnitude of the order parameter  
 280 are all proved invariant under the change of frame, so nothing observable  
 281 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on  
 282 two oscillators with a common frequency, discharging every hypothesis. The  
 283 restriction is real and we state it plainly: the reduction is exact only for  
 284 identical frequencies. A genuine spread of natural frequencies leaves residual  
 285 detunings in the reduced system, phase-locking then requires coupling above

286 a critical  $K_c$ , and that  $K_c$  — Kuramoto’s threshold for a spread of frequen-  
 287 cies, not the noise threshold  $2D$  of Section 10 — is nowhere in our Lean  
 288 development. It is studied numerically, in Figure 5, as Table S1 records.

289 A fourth defect was dimensional, and it concerned the one row of Table S1  
 290 that is still numerical rather than proved. `Phase8_ContinuousField.lean`  
 291 defines the critical coupling  $K_c = 2D$  and a predicate for being above it.  
 292 That predicate compared  $\int_x \int_y K(x, y)$  against  $2D$  — but a double integral  
 293 against an unnormalized measure carries the substrate’s mass twice over, so  
 294 a constant kernel  $K$  on a substrate of total mass  $m$  contributed  $Km^2$ . The  
 295 comparison therefore depended on how large the substrate was rather than  
 296 on how strongly it was coupled, and any kernel whatever could be pushed  
 297 above threshold by inflating  $m$ . `Examples.lean` §9 exhibits the failure con-  
 298 cretely: on a two-site substrate carrying counting measure, a system coupled  
 299 at  $K = 1/2$  against a threshold of  $K_c = 1$  — a factor of two *below* threshold  
 300 — has an unnormalized double integral of  $2 > 1$ , and would satisfy any pred-  
 301 icate stated with that integral. The fix is to require the substrate’s measure  
 302 to be a probability measure, which is the normalization in which the mean-  
 303 field derivation of  $K_c = 2D$  is carried out; `mean_field_coupling_const`  
 304 then gives that a constant kernel has mean-field strength exactly  $K$ , and  
 305 `exhibits_phase_transition_const_iff` reduces the predicate to  $K > 2D$ .  
 306 We flag what this does and does not do. It makes the predicate say something  
 307 whose truth depends only on the physics, and §9 witnesses it on both sides of  
 308 the threshold. It does not by itself prove  $K_c = 2D$ : the content of that value

309 is the bifurcation of the self-consistency equation  $r = I_1(Kr/D)/I_0(Kr/D)$   
 310 at  $K = 2D$ , which is a separate question from whether the predicate is di-  
 311 mensionally sound. That bifurcation is proved below, in both directions,  
 312 from an assumed stationary density; deriving the density itself needs the  
 313 Fokker–Planck operator and is not done.

314 We draw no grand moral from this beyond the obvious one. The value  
 315 of a proof assistant here was not that it certified our argument, but that it  
 316 made a specific, checkable claim about what had been assumed — and that  
 317 claim turned out to be false in a way no amount of reading the prose would  
 318 have revealed.

### 319 **3 Axiom 1: Symmetries as Setting, and the** 320 **Capacity of a Finite Phase Space**

321 Our starting premise relies on fundamental geometry. In our Lean formaliza-  
 322 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the  
 323 Poincaré group) act upon field states, and the invariance of the system’s total  
 324 energy under that action is recorded by the class `SymmetryInvariantAction`.  
 325 Physically, it is such an invariance that Noether’s theorem converts into the  
 326 conservation of energy and momentum. The scope of the *formal* claim is nar-  
 327 rower and we state it plainly: the invariance is a class field, i.e. an assumption  
 328 a model must discharge, and no conserved quantity is ever constructed from  
 329 it. There is no Noether theorem in our Lean, and nothing downstream of this

330 section depends on one. The class is now inhabited — `Examples.lean` §11  
 331 exhibits a double-well potential  $V(v) = (v^2 - 1)^2$  on a two-point spacetime  
 332 with the  $\mathbb{Z}_2$  action  $v \mapsto \pm v$ , discharges the invariance because  $(\pm v)^2 = v^2$ ,  
 333 and proves the vacuum manifold  $\{-1, 1\}$  degenerate with the two minima ex-  
 334 changed by the symmetry, so an energy-minimising field and its image have  
 335 equal energy and differ. That is symmetry breaking exhibited rather than  
 336 assumed; it is still not a conserved quantity, and  $\mathbb{Z}_2$  being discrete there is  
 337 no one-parameter family to differentiate along in the first place. Simultane-  
 338 ously, discrete permutation symmetries dictate the statistical mechanics of  
 339 any physical system, defining a strictly finite volume of phase space.<sup>1</sup>

340 Because phase space is finite, any localized physical system possesses a  
 341 mathematical limit on the amount of information (distinguishable states) it  
 342 can embody at any given moment. That sentence is the section’s contribution  
 343 to the chain, and three theorems carry it, in `Phase1_PhaseSpaceCapacity.lean`,  
 344 whose only hypothesis about the system is that its state space is finite: no  
 345 symmetry group, no metric, no action functional appears anywhere in them.

346 The first is the capacity bound itself. A probability distribution  $p$  on a

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<sup>1</sup>While the Poincaré group and discrete permutations serve as accessible physical ex-  
 amples, the underlying Lean formalization does not hardcode specific physics. It defines  
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-  
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-  
 ics). What the formalization supplies is the vocabulary in which symmetries, conserved  
 quantities and phase-space constraints can be *stated*; it does not derive them. All three  
 of the classes involved are now inhabited in `Examples.lean` — the  $\mathbb{Z}_2$  action and its in-  
 variance in §11, the real line with the Euclidean metric in §12 — but the instances are  
 witnesses of inhabitability, not of physics: no theorem in the development consumes a  
 symmetry group or a metric, and that half of the premise is setting rather than content.  
 The other half of the premise is not setting, and the paragraphs below are what it does.

347 finite state space  $X$  satisfies  $H(p) = -\sum_x p(x) \log p(x) \leq \log |X|$  (`shannon_`  
 348 `entropy_le_log_card`), with equality if and only if  $p$  is uniform (`shannon_`  
 349 `entropy_eq_log_card_iff`). The “only if” half is the part worth stating: it  
 350 says  $\log |X|$  is the exact capacity rather than a convenient over-estimate, so  
 351 a system holding any distribution other than the flat one over its states holds  
 352 strictly less than its size suggests. The proof is Gibbs’ inequality against the  
 353 uniform law —  $\log x \leq x - 1$  applied at  $x = 1/(|X|p(x))$ , with the  $p(x) = 0$   
 354 case carrying the slack that makes the equality condition sharp — and it is  
 355 elementary. What was missing here was the statement, not the difficulty.

356 The second says the limit binds. Let the boundary absorb a stream of  
 357 perturbations drawn from an alphabet  $P$ ; a stream of length  $k$  has  $|P|^k$   
 358 possible histories. Once  $|X| < |P|^k$ , the map from histories to final states  
 359 cannot be injective (`absorb_not_injective`), so two distinct histories leave  
 360 the system in the same state, and the entropy of the source strictly exceeds  
 361 anything the system can hold (`source_entropy_exceeds_capacity`). The  
 362 update map is arbitrary and is allowed to be reversible at every individual  
 363 step; the collision is forced by counting alone, which is the sense in which the  
 364 premise, and not the model of the dynamics, is doing the work.

365 The third is what Derivation 2 consumes, and it is the one place where  
 366 finiteness does something no other hypothesis could. On a finite state space,  
 367 a map that fails to reach every state is not injective (`is_erasure_of_not_`  
 368 `surjective`) — which is exactly the erasure condition Landauer’s bound re-  
 369 quires — so any update of a finite phase space leaving some state unreachable

370 dissipates strictly positive heat (`finite_phase_space_dissipates`). On an  
 371 infinite phase space the step simply fails: the successor map on  $\mathbb{N}$  misses 0 and  
 372 is injective. `Examples.lean` §19 records that failure next to the witnesses, so  
 373 the finiteness hypothesis is fenced rather than assumed to be load-bearing,  
 374 and runs all three results on a single bit: the fair coin attains  $\log 2$ , a  $3/4$ –  
 375  $1/4$  bit falls strictly short, two binary perturbations already outrun a one-bit  
 376 memory, and §1’s one-bit thermodynamic system pays heat for a state it  
 377 cannot reach.

378     None of this derives the finiteness of phase space. Finiteness is assumed,  
 379 as it is in any statistical-mechanical treatment, and the Poincaré group re-  
 380 mains an illustration in prose. What has been removed is the gap between  
 381 assuming the premise and using it.

## 382   4   Derivation 1: Symmetry Breaking and the 383       Inevitability of Boundaries

384 A perfectly symmetric universe is the uniform vacuum; no distinct entities  
 385 exist. For any dynamic process to occur, the continuous symmetries of the  
 386 background field must undergo spontaneous symmetry breaking.

387     When a field drops to a lower, asymmetric energy state, topology dic-  
 388 tates the natural creation of defects (such as domain walls or monopoles).  
 389 This constraint necessitates the creation of a physical boundary—a localized  
 390 topological distinction between an “inside” and an “outside.” While such

391 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate  
 392 entity, it is not sufficient for agency. The boundary is relational, defined by  
 393 the twisted geometry of the field surrounding it, and must be coupled with  
 394 non-equilibrium thermodynamics to exhibit active behaviors.

395 That first sentence is a claim about topology, and it needs a different  
 396 argument from the energy claim beside it. The energy half — an energy-  
 397 minimizing field sits in the vacuum manifold almost everywhere (`pointwise_`  
 398 `vacuum_of_global_min`) — together with the observation that a boundary  
 399 defect *obstructs* extension over a contractible interior, produces no defect:  
 400 both are statements about a field already given. What produces one is  
 401 `exists_notMem_of_connectedComponentIn`: if the substrate is connected  
 402 and a continuous field takes values in two *different connected components*  
 403 of the vacuum manifold  $M$ , then somewhere the field is not in  $M$  at all.  
 404 That excluded region is the wall. This is the  $\pi_0$  case of Kibble’s classifica-  
 405 tion of topological defects by the homotopy of the vacuum manifold (Kibble,  
 406 1976), and the proof is elementary in exactly the way it is elementary in the  
 407 physics literature: the continuous image of a connected space is connected, a  
 408 connected subset of  $M$  lies inside one component of  $M$ , and the field’s range  
 409 would be such a subset if it never left  $M$ . No hypothesis mentions energy, a  
 410 potential, or a symmetry; connectedness of the substrate and disconnected-  
 411 ness of the vacuum are the whole of it.

412 We formalize the  $\pi_0$  case and stop there. The higher cases —  $\pi_1 \neq 0$   
 413 forcing vortex lines,  $\pi_2 \neq 0$  forcing monopoles — are the rest of Kibble’s

414 classification and Mathlib’s homotopy-group library would support at least  
 415 the next one; they are available and not taken, because the argument this  
 416 paper runs uses domain walls and nothing else. The witness is the double  
 417 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be  
 418 exactly  $\{-1, 1\}$  (`wellVacuum_eq`) rather than merely known to contain both  
 419 minima, and its disconnection is proved in the form the theorem consumes:  
 420 no preconnected subset of it contains both, since a preconnected subset of  $\mathbb{R}$   
 421 is an interval and an interval spanning  $-1$  and  $1$  contains  $0$ , which is the top  
 422 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field  
 423 on *any* connected substrate that reaches both minima, with no formula for  
 424 the field anywhere in the statement. That the class is non-empty is checked  
 425 separately, because a theorem quantified over all fields is worthless if none  
 426 satisfies its hypotheses: the clipped identity on  $\mathbb{R}$  does, and the wall it is  
 427 forced to have sits at the origin (`kink_zero_notMem`).

## 428 **5 Derivation 2: The Thermodynamic Con-** 429 **straint**

430 This newly formed topological boundary is subjected to gauge perturbations  
 431 from the “outside” environment. Because the internal phase space is finite  
 432 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations  
 433 and must overwrite old internal states.

434 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-

435 formation irreversibly decreases internal entropy, mandating that the sys-  
 436 tem exhaust heat into the environment. Philosophers of physics (such as  
 437 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-  
 438 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure  
 439 (a many-to-one mapping in abstract computational states) with *thermody-*  
 440 *namic* erasure (a many-to-one mapping in physical phase space). They argue  
 441 that logical erasure could conceptually be implemented by an injective phys-  
 442 ical process without dissipating heat.

443 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)  
 444 addresses this critique by defining erasure as a non-injective mapping on the  
 445 *physical* state space ( $\neg \text{Function.Injective } t$ ). We formalize the global evolu-  
 446 tion of the bipartite universe (System + Bath) as strictly injective, repre-  
 447 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied  
 448 to abstract logical erasure, but is a mandatory mathematical consequence  
 449 of any local physical state space contraction under globally reversible dy-  
 450 namics. If a cognitive system’s internal topology forces a contraction of its  
 451 physical state space to accommodate structural resonance, heat dissipation  
 452 becomes a thermodynamic requirement. The finiteness premise enters at ex-  
 453 actly this point. What a modeller can usually check about a physical update  
 454 is not that it is many-to-one but that some state has become unreachable;  
 455 on a finite phase space those are the same condition, and `finite_phase_`  
 456 `space_dissipates` is the resulting statement. That identification preserves  
 457 the Norton–Shenker distinction rather than blurring it: an update that keeps

458 the incoming perturbation on the record is injective and dissipates nothing  
 459 (`keepRecord_injective`), while one that refreshes the register holding it is  
 460 not (`absorbStep_is_erasure`). Losing *which* history occurred is forced by  
 461 counting alone and is free; overwriting the register that held it is what is  
 462 charged  $k_B T \ln 2$ . A stable boundary thus constitutes a dissipative structure  
 463 (Prigogine & Stengers, 1984), operating as a non-equilibrium steady state  
 464 that consumes free energy to pay the thermodynamic cost of erasure. Contin-  
 465 uous thermal fluctuations play a constructive role in establishing boundaries  
 466 between states while contributing to the system’s flexible, adaptive response.

## 467 **6 Derivation 3: Prediction as the Thermody-** 468 **namic Cost of Memory**

469 A dissipative structure is driven by its environment. Write  $X_t$  for the state  
 470 of the structure and  $S_t$  for the environmental signal driving it, and let the  
 471 signal evolve by a dynamics of its own,  $S_t \rightarrow S_{t+1}$ , on which the structure  
 472 does not act. Two quantities are then defined at each step: the *memory*  
 473  $I(X_t; S_t)$ , the mutual information between the structure’s present state and  
 474 the signal’s present value, and the *predictive information*  $I(X_t; S_{t+1})$ , which  
 475 is what that same state says about the signal’s next value.

476 **What the entropy-production bound does not give.** There is a nat-  
 477 ural route from entropy production to predictive processing, and it does

478 not work. Model the environment as a stochastic driving force  $P(\text{ext})$  and  
 479 the structure's internal transitions as  $Q(\text{int})$ . Entropy production is then  
 480 bounded below by  $D_{KL}(P||Q)/\Delta t$  — this is Gibbs' inequality, it is carried  
 481 in Lean as an instance obligation, and §3 of the supplement states it. Re-  
 482 arranged it reads  $D_{KL}(P||Q) \leq \Delta t \sigma$ , so a system descending on  $\sigma$  lowers  
 483 a *bound* on the divergence, and it is tempting to conclude that the internal  
 484 transition probabilities come to mirror the environment's statistical symme-  
 485 tries. They do not, because a driven system does not relax to  $\sigma = 0$ . It  
 486 relaxes to a non-equilibrium steady state at strictly positive entropy produc-  
 487 tion, and the bound then yields only  $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$ , a positive number.  
 488 The inequality is true and does real work — combined with Gibbs it gives  
 489  $\sigma \geq 0$  — but no limit may be read off it, and a framework that wants predic-  
 490 tive processing has to get it from somewhere else. It gets it from the bound  
 491 below.

492 **Where predictive processing comes from: a published bound.** Still,  
 493 Sivak, Bell and Crooks (Still et al., 2012) prove that for a system driven by  
 494 an environmental signal the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

495 The bracket is the *nonpredictive* information: what the structure retains  
 496 about the signal's present that says nothing about the signal's future. Dis-  
 497 sipation is bounded below by it. Memory that fails to predict is not merely

498 useless to the structure; it is thermodynamic waste, and the waste is quan-  
499 tified in energy.

500 **Three consequences, and they are theorems rather than readings.**

501 The bound is carried in Lean as a class field, `PredictiveDissipation`.  
502 `still_bound (Phase3_PredictiveThermodynamics)`, for the same reason ev-  
503 ery other physical postulate here is: its `axiom` form, quantified over all joint  
504 laws with the dissipated work fixed, is refutable, and `Examples §18.4` refutes  
505 it. What is *derived* is everything downstream.

506 First, *the bracket is non-negative*, and this is proved rather than assumed.  
507 If the structure does not act on its environment, then  $X_t \rightarrow S_t \rightarrow S_{t+1}$   
508 is a Markov chain and the data processing inequality gives  $I(X_t; S_{t+1}) \leq$   
509  $I(X_t; S_t)$ : a system cannot know more about the signal's future than about its  
510 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback-  
511 Leibler divergence but no mutual information, so the latter is defined here  
512 as the divergence of the joint law from the product of its marginals, which is  
513 the standard definition and makes the library's data processing lemma apply  
514 directly.

515 Second, *the second law follows* instead of being assumed. Dissipation is  
516 bounded below by a quantity now known to be non-negative, so  $W_{\text{diss}} \geq 0$   
517 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays  
518 for the entropy-production bound, with one difference that matters: Gibbs'  
519 inequality compares a system to an arbitrary reference distribution, whereas

520 the data processing inequality compares the system to its own future, which  
 521 is a quantity selection can act on.

522 Third, and this is the point of the replacement, *the inference now runs*  
 523 *in the valid direction*. The quantity of interest sits on the small side of  
 524 the inequality:  $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$ . No limit is required.  
 525 Any bound on dissipation is immediately a bound on wasted memory, so a  
 526 structure held to an energy budget is thereby held to a prediction standard:  
 527 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case  
 528  $W_{\text{diss}} = 0$ , the theorem `predictive_eq_of_no_dissipation` gives that every  
 529 retained bit is predictive. This is a weaker statement than the one it replaces,  
 530 and it is the one that is true.

531 **The hypothesis is load-bearing, and we exhibit what breaks with-**  
 532 **out it.** The data processing step requires that the structure not write to  
 533 its environment. In the formalization this is not an assumption stated beside  
 534 the theorem but the shape of the evolution itself: the signal's step is applied  
 535 to the pair as the parallel kernel  $\text{id} \parallel \kappa$ , which acts as the identity on the  
 536 system coordinate and through  $\kappa$  on the signal coordinate, so no channel  
 537 from  $X_t$  to  $S_{t+1}$  exists. **Examples** §18.3 shows the restriction is not decora-  
 538 tion: a Markov kernel on the *pair* which copies the system's state into the  
 539 signal carries an independent two-bit law to a perfectly correlated one, rais-  
 540 ing mutual information from zero to a positive value. An agent that steers  
 541 its environment can indeed know more about that environment's future than

542 about its present, and Still’s bound is not about such an agent.

543 **Both regimes are exhibited, not just described.** Examples §18 builds  
544 two systems over one correlated two-bit joint law, differing only in the signal’s  
545 dynamics. When the signal is frozen, every remembered bit is a predictive  
546 bit, the nonpredictive information is zero, and the instance is permitted to  
547 dissipate nothing — with the memory itself proved strictly positive, so the  
548 conclusion is about a system that does remember something. When the signal  
549 is redrawn from its stationary law at each step, the predictive information is  
550 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*  
551 to dissipate  $k_B T$  times its whole mutual information. Establishing that the  
552 mutual information of the witness is neither zero nor infinite is where the  
553 work of that section lies.

554 **An equality, where the entropy-production route had only a bound.**  
555 Kawai, Parrondo and Van den Broeck (Kawai et al., 2007) show that the  
556 average dissipated work is exactly  $k_B T$  times the relative entropy between the  
557 forward and time-reversed phase-space densities. Carried here as a predicate  
558 on an instance rather than as a further postulate, it replaces the abstract  
559 dissipation field with a quantity an experiment can estimate — the statistical  
560 distinguishability of a recording of the system from the same recording run  
561 backwards — and yields the statement that the irreversibility of the dynamics  
562 is at least the amount of memory that fails to earn its keep (`nonpredictive_`  
563 `le_arrow`).

564 **What this does not establish.** It does not deliver the Free Energy Prin-  
565 ciple. Still’s bound says that prediction is thermodynamically favoured; it  
566 says nothing about hierarchical generative models, variational free energy,  
567 or Bayesian inference, and we do not claim to have derived any of them.  
568 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,  
569 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-  
570 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is  
571 proved here, which is a weaker relation than derivation and the only one  
572 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).  
573 Maximal predictive information is compatible with the structure storing a  
574 lossy, compressed or otherwise unrecognisable function of the signal; a state  
575 that predicts is not thereby a state that means. The narrower claim is the  
576 one worth making: whatever internal states a physical structure holds, those  
577 that do not help predict its environment are paid for in heat, so the pressure  
578 toward states indexed to the environment is energetic rather than semantic.  
579 We continue to call the phenomenon *structural resonance*, and there is no  
580 teleological drive to “survive” in it: predicting the environment is simply the  
581 cheaper of the two ways to hold a memory.

## 582 7 Macroscopic Scaling: A Theorem and a 583 Commitment

584 This is the chain’s fourth step, and it is deliberately not numbered as a  
585 derivation. Three steps of different kinds meet here and are easy to run to-  
586 gether: convergence of a scalar energy, which is a theorem; introduction of a  
587 continuum coupling kernel, which is a modelling assumption; and identifica-  
588 tion of that kernel with the brain’s endogenous electromagnetic field, which  
589 is an empirical claim about cortex. Neither later step inherits the licence of  
590 the first. Kept apart, the empirical claim is the framework’s central commit-  
591 ment and the place where it is most easily wrong — which is where a claim  
592 of this kind belongs.

### 593 The formal half: coarse-graining is a theorem, and a 594 narrow one

595 A network of discretely coupled units has a coupling energy that is a sym-  
596 metric double sum over pairs. `Phase2.MeshConvergence.lean` proves that  
597 this sum converges to a continuous integral: along a sequence of regular  
598 triangulations of a region  $S$  whose edge regions shrink to points, the dis-  
599 crete energy  $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$  converges to  $\int_S f d\mu$  (`mesh_refinement_`  
600 `convergence`), witnessed on the uniform partition of  $[0, 1)$  in `Examples §6`.  
601 The identification that makes this a coarse-graining rather than an arbitrary

602 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,  
603 the double sum is exactly the midpoint rule over unordered edges, so the  
604 discrete energy is a quadrature of the continuous one rather than a different  
605 quantity that happens to converge to it.

606 What this establishes is worth stating exactly, because it is less than the  
607 phrase “coarse-graining, governed by the Renormalization Group” suggests.  
608 It is a statement about an *energy functional*, not about a dynamics: nothing  
609 here shows that trajectories of the discrete Kuramoto system converge to so-  
610 lutions of a field equation, and no renormalization-group flow is constructed.  
611 The theorem converges one real number, the energy; it does not produce a  
612 scalar density, an electromagnetic field or a kernel of two continuum points.  
613 The continuous object used later, `ContinuousNeuralField`, is *posited* with  
614 a coupling kernel  $K$ . A genuine kernel limit first needs new discrete architec-  
615 ture over  $M \times M$ , since `edge_region` supplies only subsets of  $M$ ; only after  
616 such a kernel exists does the separate propagation-of-chaos problem for the  
617 dynamics arise.

## 618 **The modelling and empirical steps: positing and iden-** 619 **tifying the kernel**

620 The formal half supplies scalar energy convergence and no kernel. The field  
621 sections therefore posit a symmetric continuum kernel as a modelling input.  
622 Which physical variable in cortex realizes that independently supplied kernel

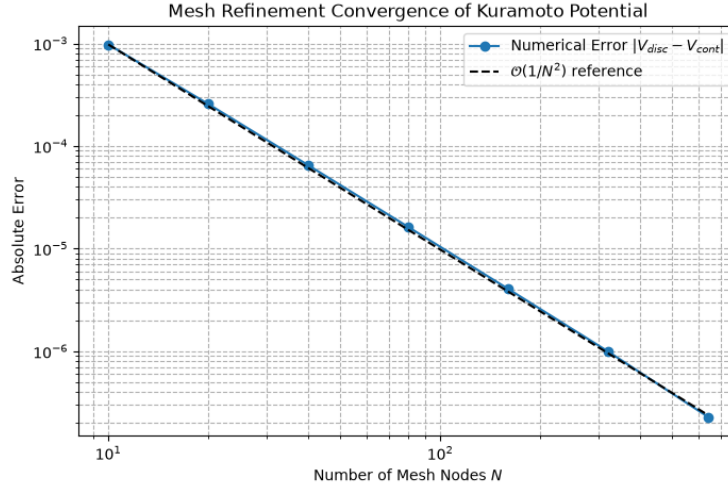


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of  $N$  units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field  $\theta(x) = \sin x$ , is compared against the continuous integral it is proved to converge to. The error falls as  $\mathcal{O}(N^{-2})$ , the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones.

623 is a further empirical question, and we state our answer as a commitment:

624       *The model identifies its posited continuum kernel with the endoge-*  
625       *nous electromagnetic field of mammalian cortex.*

626       This is a claim about biology, not a consequence of anything above it.  
627       It is what makes the framework falsifiable and it is what we expect to be  
628       attacked.

629 **What supports it.** Neurons do not only signal through synapses: trans-  
630 membrane currents generate extracellular fields that shift the membrane po-  
631 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-  
632 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike  
633 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic  
634 head models put mesoscopic ephaptic effects within the range where they  
635 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic  
636 treatments show field coupling is sufficient for coincidence detection in the  
637 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have  
638 been argued to carry information that survives representational drift in the  
639 underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha &  
640 Lima, 2024). The field is spatially continuous and superposes contributions  
641 from large populations, which is the property the framework needs: it is a  
642 medium in which spatially separated activity is combined without a further  
643 mechanism for combining it. This is the integrative role the CEMI field  
644 theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

645 **What the evidence supports, and what it does not.** The endoge-  
646 nous field’s effect on any individual neuron is *modulatory*. Anastassiou and  
647 Koch report that endogenous fields of 1–5 mV/mm shift spike timing by one  
648 to three milliseconds (Anastassiou & Koch, 2015), an effect that is real, re-  
649 producible, and subthreshold: it biases when a cell fires; it does not decide  
650 whether it fires. Nothing in the work cited above supports calling the field

651 the *primary* dissipative structure of the system, and we do not claim it. The  
652 claim is that the field is *modulatory in its effect on individual neurons and*  
653 *integrative in its spatial reach* — weak per synapse, and the only variable in  
654 the system that is defined everywhere at once. What the framework needs  
655 is the reach, not the strength, and the two are easy to conflate.

656 **Why reach is the property that matters.** A theory of consciousness  
657 needs an account of why many local processes make one experience, and most  
658 supply it by stipulation or by a measure. The account here is structural, and  
659 it is the framework’s strongest argument. The field is the only variable in the  
660 system that is *defined everywhere at once*. Unity is not asserted of it; unity  
661 is what a field is. That is what makes Derivation 5’s sheaf-theoretic step  
662 available at all — local states glue to one global section because there is one  
663 object over which gluing can happen — so the answer to the combination  
664 problem is a piece of geometry rather than a postulate. Compare “integration” in Integrated Information Theory, which is a number, or “broadcast” in  
665 Global Workspace Theory, which is a metaphor about a workspace. It is also  
666 what the empirical identification above is carrying: if the posited continuum  
667 coupling is realized by something that is *not* defined everywhere at once —  
668 by synaptic connectivity, say — then this argument is unavailable, and the  
669 framework loses the property that distinguishes it from its rivals rather than  
670 merely losing a mechanism.

672 **Its critics.** The objection with the longest history is that the fields are  
673 too weak to matter, and it has been pressed from within the field-theoretic  
674 camp: Pockett, who has herself defended an electromagnetic theory of con-  
675 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the  
676 measurement problem of relating gross field structure to reported experience,  
677 the absence of a constant relationship between the two, and the question of  
678 whether the field does causal work or merely reflects it. The quantitative case  
679 for scepticism has strengthened since. Vöröslakos and colleagues measured  
680 intracranial fields during transcranial stimulation in rats and human cadav-  
681 ers and found that the great majority of applied current is shunted through  
682 the scalp, so that conventional protocols deliver fields at or below the 1 V/m  
683 range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields  
684 rather than endogenous ones, but it constrains how much causal work a field  
685 of that magnitude can be doing, and it is the reason we do not claim the  
686 field drives spiking.

687 **What would falsify it.** We state the failure conditions rather than leaving  
688 them to a referee.

689 1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted  
690 — by a conductive medium that nulls the extracellular potential with-  
691 out altering transmembrane currents — and neither behaviour nor re-  
692 port changes, the identification fails. This is the decisive experiment  
693 and it is, in principle, available.

694 2. *A magnitude gap.* If the spatial integration the framework requires  
695 demands field effects an order of magnitude above the 1–5 mV/mm  
696 measured in cortex, the commitment fails on arithmetic. The relevant  
697 number is not whether ephaptic coupling exists but whether its coupling  
698 strength suffices to place the system above the critical coupling  $K_c$  of  
699 Derivation 7.

700 3. *A timescale mismatch in the wrong direction.* The prediction of Sec-  
701 tion 11 turns on the field’s dynamics being fast relative to the astrocytic  
702 and clearance variables. If recovery from sleep inertia tracks an expo-  
703 nential set by the electrical timescale rather than the delayed sigmoid  
704 the slow variable predicts, the framework loses the one prediction that  
705 is its own.

706 4. *Dissociation of field coherence from reported unity.* If large-scale field  
707 coherence is preserved across states in which experiential unity is lost  
708 — or absent in states where it is retained — the identification of the  
709 global section with unified experience has no physical carrier.

710 A framework that survives all four is not thereby true. It is merely still  
711 standing, which is more than most theories of consciousness arrange to be  
712 able to lose.

713 **Geometric frustration, and the regime our theorems ac-**  
714 **tually cover**

715 The system’s memory capacity is standardly grounded in *geometric frustra-*  
716 *tion*: a highly coupled network cannot bring all its nodes to their lowest en-  
717 ergy simultaneously, producing a jagged landscape of near-degenerate states  
718 with the capacity to store many patterns. That argument is standard and it  
719 is not what this development formalizes. Confronting the discrepancy is the  
720 honest course, so we do it here rather than in a footnote.

721 Frustration requires competing couplings — in the Kuramoto setting,  
722 a sign structure that no phase assignment can satisfy on every edge at  
723 once. The formal setting admits this: `KuramotoSystem` carries  $A : V \rightarrow$   
724  $V \rightarrow \mathbb{R}$ , symmetric and otherwise unconstrained, so signed couplings are  
725 representable. The question is therefore not whether the development can  
726 speak about a frustrated network but *where along the chain* the sign of  $A$   
727 is constrained, and that question has an exact answer. Positivity is not a  
728 hypothesis distributed through the development; it is a *field of one class*.  
729 `ThermodynamicCover` (`Phase5_GlobalSection`) demands `A_pos` of every in-  
730 stance, so it is demanded at the node where local sections are glued and  
731 at the edge that builds a cover, and it is demanded nowhere upstream of  
732 that node. The theorems that feed the class carry it with them — `phase_`  
733 `locked_minimizes_potential` assumes  $A_{ij} > 0$ , and the characterization of  
734 the minimizers as the phase-locked configurations inherits it, while the bridge

of `Phase5_EquilibriumBridge` and the convergence theorem of `Phase4_RotatingFrame` §7 require the strictly stronger uniform bound  $0 < a \leq A_{ij}$ , since  $a$  enters the Łojasiewicz constant. A network with all-positive couplings is an unfrustrated ferromagnet whose potential has one minimum up to a global phase shift. It is the opposite of a jagged landscape, and it has no capacity to store multiple patterns at all.

Locating the constraint that precisely narrows the objection, and the narrower version is the one we accept. It is not that the framework describes an unfrustrated brain throughout; the mean-field reduction of Derivation 7 has no sign structure to be frustrated at all, since its coupling is a scalar  $K$  rather than a matrix. It is that **one step** of the framework — the gluing of Derivation 5 — describes an unfrustrated system, and every step before it does not care.

The two halves are worth listing by name, because the surviving half is larger than the concession usually made. What holds with no assumption whatever on the sign of the coupling is the entire dissipative account: well-posedness on all of  $\mathbb{R}$  (`is_kuramoto_trajectory_exists`, `_unique`); the Lyapunov identity itself, that the potential’s time derivative is minus the summed squared velocity (`dV_dt_le_zero`), which is the theorem the whole descent argument rests on and which mentions no positivity hypothesis; covariance under passage to a rotating frame; monotone descent of the potential (`dynamic_potential_antitone`) and its convergence to a finite limit (`dynamic_potential_tendsto`); finiteness of the dissipation integral; and —

758 through Barbalat’s lemma — that the motion stops along every trajectory  
 759 (`velocity_sq_tendsto_zero`). Every one of these is stated for an arbitrary  
 760 symmetric  $A$ , so a frustrated network runs, dissipates, and comes to rest  
 761 inside this development exactly as an unfrustrated one does. What requires  
 762 positivity is the identification of the resting place: the three minimization  
 763 results of `Phase4_KuramotoDynamics`, which between them characterize the  
 764 potential’s minimizers as exactly the phase-locked configurations; the class  
 765 field `ThermodynamicCover.A_pos`; and the convergence theorem `kuramoto_`  
 766 `tendsto_global_minimum`. *That* the motion stops is frustration-agnostic;  
 767 *where* it stops is not, and the claims this framework makes about the des-  
 768 tination all live on the positive side of that line. On the continuum side  
 769 the accounting differs slightly: `phase_locked_minimizes_entropy_of_symm`  
 770 assumes only that the kernel is symmetric and so tolerates signed kernels,  
 771 but it is conditional on a dynamically phase-locked field existing — which is  
 772 precisely what frustration may prevent.

773 We therefore scope the claim rather than repeat it. The memory capacity  
 774 of a frustrated cortical network is not supported by any theorem here, and we  
 775 do not assert it as part of the chain. `Phase5_TwistedGluing` tests the natural  
 776 repair: let local states agree on overlaps up to a phase and ask whether  
 777 their offset cocycle is a coboundary. The result is negative in the physically  
 778 immediate case. Locked, twisted and splay configurations all assign a phase  
 779  $\phi_i$  to every patch, hence their offsets are  $\phi_i - \phi_j$ , automatically a coboundary  
 780 (`isCoboundary_of_phaseField`); the family therefore glues after patchwise

781 rotation (`gluesUpToPhase_of_phaseField`). Frustration alone supplies no  
782 non-trivial obstruction and hence no content label.

783 There is, however, a measured cortical quantity with the right topolog-  
784 ical shape to motivate a different construction. Primate LFP phase maps  
785 contain phase singularities whose surrounding electrode loops have integer  
786 winding number; these complex waves covary with local spiking activity  
787 (Townsend et al., 2015). Human cortical spiral centres and rotation direc-  
788 tions are also task-dependent and classify cognitive tasks (Xu et al., 2023).  
789 The observable is the winding of a circle-valued phase around a point where  
790 its amplitude vanishes, not an independently chosen circle-valued offset be-  
791 tween patches. Its natural transition data are the integer multiples of  $2\pi$   
792 relating *local real-valued lifts* of phase on overlaps. `Phase5.PhaseLifts` for-  
793 malizes the measured loop at this discrete level: `loopWinding_eq_zero_of_`  
794 `hasGlobalLift` proves that transitions from one global real lift telescope to  
795 zero, while `threeElectrodeWinding_not_hasGlobalLift` witnesses a three-  
796 patch loop with winding one and no global lift. This is not the earlier  
797 `PhaseObstruction`, whose circle-valued offsets from absolute patch phases  
798 necessarily telescope.

799 The ambient object and its measurement map can now be fixed without  
800 manufacturing a full Čech complex. For a cortical surface  $M$  and band-  
801 limited analytic field  $z(x, t) = A(x, t)e^{i\phi(x, t)}$ , let  $Z_t$  be the preregistered set  
802 of amplitude-zero phase singularities and let  $M_t^* = M \setminus Z_t$ . The measure-  
803 ment map is the normalized phase  $q_t : M_t^* \rightarrow S^1$ ,  $q_t = z/|z|$ . An electrode or

804 source-space cover  $\{U_i\}$  of  $M_t^*$  is chosen independently of trial outcome, with  
 805 local real lifts  $\tilde{\phi}_i$  of  $q_t$ ; on a connected overlap,  $(\tilde{\phi}_i - \tilde{\phi}_j)/(2\pi)$  is the measured  
 806 integer transition. Loops in the cover nerve therefore feed `loopWinding` di-  
 807 rectly. In sampled data, “zero” must be an amplitude threshold fixed from  
 808 held-out baseline data, and a singularity is accepted only when a surround-  
 809 ing loop has stable non-zero integer winding across prespecified spatial and  
 810 temporal scales.

811 This construction supplies a falsifiable content test, not a content identifi-  
 812 cation. On trials with an independently recorded content label (for example,  
 813 percept reports under ambiguous or matched stimulation), compare a control  
 814 model using stimulus, task, response, eye movement and arousal covariates  
 815 with a nested model that additionally uses signed winding number, source-  
 816 localized singularity position, rotation direction and lifetime. Model choice  
 817 and all thresholds are preregistered; prediction is evaluated on held-out par-  
 818 ticipants and sessions, and label permutations are restricted within task,  
 819 stimulus and arousal strata. The candidate survives only if the topological  
 820 variables improve out-of-sample discrimination of content after controlling  
 821 for task and arousal, and if that increment replicates when motor report is  
 822 changed or omitted. Failure to improve the control model rejects the pro-  
 823 posed content role. Success would establish an empirical association beyond  
 824 those controls, not that a winding sector is phenomenal content, and this  
 825 branch remains outside the conditional chain.

## 826    **The supporting physiology**

827    The continuous medium is physically supported by astrocytic syncytia—non-  
828    neuronal networks joined by gap junctions that provide macroscopic cytoplas-  
829    mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).  
830    Managing the thermodynamic cost of this state requires sophisticated micro-  
831    bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-  
832    tochondrial Information Processing Systems (MIPS), transducing biological  
833    and environmental information to sustain the coherence of the macroscopic  
834    field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled with mito-  
835    chondrial micro-mechanics, generate and sustain the field. Because the exter-  
836    nal environment is continuous, achieving maximal thermodynamic structural  
837    resonance is optimally realized within this continuous macroscopic medium  
838    (van Bree et al., 2025).

## 839    **8    Derivation 5: Global Sections and the Unity** 840    **of Consciousness**

841    The system is now characterized by a continuous macroscopic electrodynamic  
842    field generated by millions of coupled micro-components, forced to resonate  
843    with the environment. Following Section 7, the coupling regime assumed  
844    from here on is the unfrustrated one — all couplings positive — which is  
845    what every theorem below requires and what the frustrated network of the

846 memory argument is not.

847     The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-  
848 tors guarantees that, past a critical coupling threshold, these distributed  
849 local fields undergo a phase transition into unified synchronization (Bauer et  
850 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this  
851 phase-locked state minimizes the Lyapunov potential of the system, in the  
852 precise sense discussed in Section 2.1: the functional minimized is the cou-  
853 pling potential read in the frame rotating with the natural frequency, which  
854 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a  
855 system of identical oscillators.

856     The phase transition supplies synchronized phases. A separate compati-  
857 bility hypothesis says that the corresponding local states agree on every over-  
858 lap; given that hypothesis, the sheaf condition mathematically glues them  
859 into a singular *Global Section* (`Phase5_GlobalSection.lean`). The EM field  
860 acts as a single, indivisible topological space, providing the *Unity of Con-*  
861 *sciousness*—a necessary condition for integrated experience. While Global  
862 Workspace Theory (Baars, 1997; Dehaene & Marti, 2014) models this unity  
863 as a broadcast architecture and Integrated Information Theory (Tononi &  
864 Koch, 2016) quantifies it via  $\Phi$ , the framework here proposes a continuous  
865 physical mechanism whose overlap condition remains a modelling input. The  
866 contrast is worth drawing carefully, because all three approaches address the  
867 same explanandum from different directions.

868     We state the scope of the Lean result precisely, because it is narrower

869 than the prose above. `ThermodynamicCover` demands that the cover's phase  
 870 field minimize the reduced Kuramoto potential, and `potential_min_iff_`  
 871 `phase_locked` characterizes those minimizers exactly: they are the phase-  
 872 locked configurations, in both directions. Every instance is therefore already  
 873 phase-locked (`ThermodynamicCover.phase_locked`), and the class's single  
 874 condition on its local data — that patches at a common phase agree where  
 875 they overlap (`section_agrees_of_phase_eq`) — makes the family of local  
 876 sections compatible. `probability_glue_unique`, which is the sheaf property  
 877 of the probability presheaf and contains no physics at all, then glues that  
 878 family into exactly one global section.

879     The shape of the class is what makes this an emergence theorem rather  
 880 than a uniqueness theorem, and the difference is easy to lose. Were `LocalSectionSynchronization`  
 881 to carry a *global* section as a class field, together with a second field declar-  
 882 ing every local section to be a restriction of it, the global object would exist  
 883 before the sheaf condition was ever consulted, and `global_section_from_`  
 884 `thermodynamics` could do no more than rule out competitors to an object  
 885 the instance had supplied. It carries neither. Every field of the class men-  
 886 tions a single patch or a single overlap; the global section is constructed  
 887 (`ThermodynamicCover.invariantMeasure`); and the statement that the lo-  
 888 cal sections are restrictions of one common measure is a *theorem* about that  
 889 construction (`ThermodynamicCover.sync_to_section_eq`), true because the  
 890 sheaf condition says so, not because an instance was required to name a  
 891 measure.

892 `Examples.lean` §14 is the witness that the difference is not notational.  
 893 Two patches carry two different mass profiles,  $(2, 1, 7)$  and  $(5, 1, 3)$  across  
 894 the three sites, agreeing only at the site they share, where both put mass 1  
 895 — which is what discharges the overlap condition, by computing two num-  
 896 bers rather than by construction. The section the theorem returns has pro-  
 897 file  $(2, 1, 3)$ : it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`  
 898 `rightW`), and neither of them restricts correctly to both patches (`leftW_`  
 899 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-  
 900 tion that no single patch had. `Examples.lean` §13, built instead through  
 901 the constructor `LocalSectionSynchronization.ofInvariantMeasure`, sits  
 902 alongside it for the negative result it carries: its phase field is not the con-  
 903 stant function — the second patch a full turn ahead of the first — and by  
 904 `ThermodynamicCover.phase_locked` that is as far from constant as any in-  
 905 stance can get, since no instance can carry phases differing by anything a  
 906 measure is allowed to see.

907 Two hypotheses carry physical content. One of them is an obligation on  
 908 every instance: that synchronized patches agree on their overlaps is a mod-  
 909 elling assumption, not a consequence of any dynamics we formalize. The  
 910 other, `thermodynamic_equilibrium`, asks that the cover already sit at the  
 911 potential minimum, and no theorem says an *arbitrary* system’s dynamics  
 912 takes it there. “A thermodynamic phase transition into unity” is therefore  
 913 doing causal work that no general Lean result supports, though not work  
 914 without any support at all: `Examples.lean` §15 exhibits a two-oscillator tra-

915 jectory that starts a quarter turn out of phase and converges to the global  
 916 minimum of the dynamic potential, with the order parameter tending to 1.  
 917 That trajectory is not an isolated construction: `kuramoto_cauchy_problem`  
 918 proves the Kuramoto initial value problem well posed on all of  $\mathbb{R}$ , so through  
 919 every initial state there passes exactly one trajectory, and the exhibited one  
 920 is the only solution through its own. Nor is its settling down peculiar to  
 921 it: along every trajectory of a zero-frequency system the dynamic potential  
 922 converges, and the total dissipation  $\int_0^\infty \sum_i \dot{\theta}_i^2$  is finite and equal to the po-  
 923 tential's total drop (`dynamic_potential_tendsto`, `dissipation_integral_`  
 924 `tendsto`). The motion also stops: a finite integral does not by itself force  
 925 its integrand to zero, but the dissipation rate is Lipschitz in time, so Bar-  
 926 balat's lemma applies, and every trajectory satisfies  $\sum_i \dot{\theta}_i^2 \rightarrow 0$  (`velocity_`  
 927 `sq_tendsto_zero`; Mathlib has no Barbalat, so it is proved here as `tendsto_`  
 928 `zero_of_lipschitz_of_integral_le`). That last step — locating where the  
 929 motion stops — is taken conditionally, and by a route that avoids the LaSalle  
 930 invariance principle, which Mathlib does not carry. LaSalle is the tool for  
 931 a system with a decreasing functional and nothing else; this one has more,  
 932 and the extra structure is what makes the conclusion reachable without it.  
 933 The flow is a gradient flow of the dynamic potential, so  $\dot{V} = -\sum_i \dot{\theta}_i^2$  is an  
 934 identity rather than an inequality; the minimisers of  $V$  are characterised ex-  
 935 actly; and on the region where every pair of phases lies within a quarter turn  
 936 a Polyak–Łojasiewicz estimate holds,  $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$ , where  $W$  is  
 937 the potential's excess above its minimum and  $a > 0$  is a lower bound for the

coupling (`lojasiewicz_estimate`). Its proof is two extremal indices and the  
 identity  $\sin^2 D = (1 - \cos D)(1 + \cos D)$ : the site carrying the largest phase  
 is pulled backwards at rate at least  $a \sin D$ , where  $D$  is the spread, while the  
 excess is at most  $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$ . The compactness objection dissolves  
 as well, because the invariant region is produced by the Lyapunov function  
 rather than assumed: a single pair contributes  $\frac{1}{2}a(1 - \cos)$  to the excess, so  
 an excess below  $a/2$  forces every pairwise cosine positive for all time, and  
 continuity on the connected half-line then prevents any pair from chang-  
 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to  
 $W(t) \leq W(0)e^{-ct}$  with  $c = 2a^2 / \sum_{ij} A_{ij}$  (`excess_decay`), the rate makes each  
 $|\dot{\theta}_i|$  exponentially small and hence integrable on  $[0, \infty)$ , and a trajectory with  
 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`  
`minimum`: if the phases start within a quarter turn of each other and the initial  
 excess is below  $a/2$ , the trajectory converges to a phase-locked configuration  
 which is a *global* minimiser of the dynamic potential, with order parameter  
 $r^2 = 1$ . That is the configuration `thermodynamic_equilibrium` postulates,  
 reached rather than assumed. `Examples.lean` §17 fires it on three sites with  
 initial data  $(0, 0, \frac{1}{2})$ , where the Kuramoto system has no closed-form solu-  
 tion: the trajectory comes from the existence theorem and its limit from the  
 convergence theorem, so neither is written down.

`Phase5.EquilibriumBridge.lean` closes the gap that remained between  
 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a  
 cover whose phase field is the limit of such a trajectory and returns an in-

961 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*  
 962 *vergence*; the cover's coupling matrix is the coupling matrix of the dynamics  
 963 rather than a separately chosen one, so the system whose equilibrium the  
 964 class asserts and the system whose trajectory is run are the same system.  
 965 `Examples.lean` §17.1 is the resulting witness: three patches over the three-  
 966 site substrate, each carrying its own mass profile on its own open set, above a  
 967 phase field that no line of the construction writes down. The trajectory that  
 968 reaches it starts at  $(0, 0, \frac{1}{2})$ , which is provably not phase-locked (`trioCover_`  
 969 `start_not_locked`), so on this cover the configuration Derivation 5 requires  
 970 is arrived at rather than posited. The state the three patches glue to has  
 971 profile  $(2, 1, 3)$  and is none of the three profiles the instance carries (`trioW_`  
 972 `ne_glued`), so one cover now carries both the emergence reading of the gluing  
 973 and the derived reading of the equilibrium. What is still assumed on it is the  
 974 overlap agreement, and nothing else.

975 **The step that carries the positivity.** `ThermodynamicCover` demands  
 976 one thing more, and it is worth naming here rather than only in the limita-  
 977 tions section, because this is the step at which it enters the chain. The class  
 978 field `A_pos` requires every coupling to be strictly positive. Nothing upstream  
 979 of this node requires it: the trajectory exists, the potential descends, the  
 980 dissipation integral is finite and the motion stops for an arbitrary symmetric  
 981 coupling matrix, signed entries included. The unfrustrated regime is there-  
 982 fore not an assumption the framework makes about the brain throughout; it

983 is an assumption Derivation 5 makes about the configuration it glues, and it  
984 is the one place a frustrated network would part company with the develop-  
985 ment. The consequences are set out under *Geometric frustration* above.

986 The two hypotheses cannot be dropped, and this is the honest scope of  
987 the result. Splay and twisted configurations are equilibria of the same flow,  
988 so “every trajectory reaches the unified state” is false as a general statement,  
989 and the initial-data condition is the standard arc condition under which the  
990 true statement holds. What is claimed is therefore conditional — unity is  
991 reached from configurations already sufficiently coherent — and not that an  
992 arbitrary initial state is carried into unity. What Lean establishes about the  
993 gluing itself is the other half: *given* overlap-compatible local patches, a global  
994 state exists, is unique, and is produced by the gluing rather than presupposed  
995 by it.

996 Integrated Information Theory begins from phenomenological axioms and  
997 derives a scalar  $\Phi$  measuring irreducibility of a system’s cause-effect structure,  
998 with the exclusion postulate selecting a single maximum- $\Phi$  complex as the  
999 locus of experience. Our sheaf-theoretic global section plays a structurally  
1000 analogous role — it too picks out a distinguished unified object — but it  
1001 is arrived at from the opposite end: not postulated from the character of  
1002 experience, but forced by a thermodynamic phase transition in a physical  
1003 field. This buys two things IIT does not have. First, the unity condition  
1004 is checkable from field dynamics rather than requiring the computation of  
1005  $\Phi$ , which is intractable for systems of realistic size and whose definition has

1006 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls  
 1007 out rather than being postulated: gluing either succeeds on a given cover or  
 1008 it does not, so there is no separate principle needed to rule out overlapping  
 1009 conscious subsystems. It also inherits IIT’s exposures. Critics have argued  
 1010 that  $\Phi$  can be high in systems with no plausible claim to experience, such  
 1011 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable  
 1012 to the mirror objection, since a physical system could in principle satisfy  
 1013 the gluing condition without being cognitively interesting. We take this  
 1014 seriously rather than dismissing it: our reply is that gluing is necessary but  
 1015 not sufficient, and that the topological boundary of Derivation 1 and the  
 1016 reflexive structure of Derivation 6 are additional required conditions. IIT, by  
 1017 contrast, treats  $\Phi$  as both necessary and sufficient.

1018 Global Workspace Theory faces a complementary difficulty. It specifies  
 1019 a functional architecture — a limited-capacity workspace broadcasting to  
 1020 specialized modules — without specifying what physical property of broad-  
 1021 casting makes the broadcast content conscious, leaving the theory open to the  
 1022 charge that it explains access consciousness while leaving phenomenal con-  
 1023 sciousness untouched (Block, 1995). Our framework can be read as supplying  
 1024 the missing physical substrate for the workspace: the continuous macroscopic  
 1025 field *is* the medium in which integration is realized, and the  $K_c = 2D$  thresh-  
 1026 old gives a physical condition for ignition rather than a functional stipulation  
 1027 of it. Where GWT is agnostic about implementation, we make a falsifiable  
 1028 commitment — unity requires a continuous physical medium, which predicts

1029 that a functionally equivalent discrete broadcast architecture would not be  
 1030 conscious. We regard this as a genuine cost of the position, not a free ad-  
 1031 vantage: it is the point at which our framework departs most sharply from  
 1032 computational functionalism and where it is most readily falsified. Contrast  
 1033 this derivation with panpsychism: consciousness is not assumed as a funda-  
 1034 mental property of matter. It requires the initial existence of a topological  
 1035 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-  
 1036 dynamic phase transition into unity that occurs when integrated spatial cou-  
 1037 pling exceeds the thermodynamic noise threshold ( $K_c = 2D$ ). Without both  
 1038 the foundational topological boundary and the subsequent mathematically  
 1039 defined phase transition, the system remains a fragmented, non-conscious  
 1040 collection of micro-states. A unified field of perception, however, is not yet  
 1041 a “Self.”

## 1042 **9 Derivation 6: Reflexive Topology and the** 1043 **Emergence of the Self**

1044 While a global section provides Unity, a “Self” requires reflexivity. The  
 1045 unified EM field is a dissipative structure that must resonate with external  
 1046 perturbations while managing its own internal phase transitions. Time evo-  
 1047 lution naturally drives the field to minimize thermodynamic friction with its  
 1048 own internal state-changes.

1049 This mandates *auto-resonance*: the field must predict itself. The “Self”

1050 emerges as a higher-order topological feature when the internal geometry  
 1051 folds to include a low-dimensional avatar of its own computational and causal  
 1052 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)  
 1053 (Levin, 2019). The system is unified, predictive, and explicitly maps its  
 1054 own distinct existence from the environment. This architecture provides the  
 1055 physical foundation for a self-aware consciousness, bounded by the continuous  
 1056 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).

1057 **What carries the weight.** `reflexive_topology_implies_self` is the  
 1058 Banach fixed-point theorem with the physical commitments in its hypotheses:  
 1059 the space of global sections is assumed nonempty, metric and complete, and  
 1060 the predictive map is assumed to contract. `Examples.lean` §10 discharges all  
 1061 four on the three-site substrate of Derivation 5, and the metric is constructed  
 1062 rather than posited. The section first pierces the sheafification: on a discrete  
 1063 substrate every point has a smallest open neighbourhood, so the mass a germ  
 1064 carries at a point is well defined, and `massEquivOn` proves that over *any* open  
 1065 the sections of the probability sheaf *are* the finite measures on that open,  
 1066 read as densities against counting measure — the global case being  $U = \top$ .  
 1067 Mathlib supplies no inverse to this sheafification — its `TopCat` sheafification  
 1068 has neither an adjunction nor an `isIso_toSheafify` — so the correspon-  
 1069 dence had to be constructed. The metric on sections is then the uniform  
 1070 distance between the densities of the measures they glue to — equivalent to  
 1071 their total variation on a finite substrate, though not equal to it — and it

1072 is proved complete by exhibiting the limit section of a Cauchy sequence of  
1073 densities, not by the degenerate route of showing Cauchy sequences are even-  
1074 tually constant. The choice of metric is load-bearing rather than incidental:  
1075 under the 0/1 metric on sections every contraction is constant (`contracting_`  
1076 `implies_const`), so a witness built there would discharge every hypothesis  
1077 of the theorem and say nothing. That theorem is kept in the source, because  
1078 which metrics make the fixed point trivial is part of what the fixed point is.

1079 **The map is the avatar, or there is no map.** A reflexive boundary could  
1080 be defined with the predictive map as free data sitting beside the avatar, and  
1081 the reason it is not is a theorem rather than a preference. With the map  
1082 free, the fixed-point theorem projects it out and never touches the avatar, so  
1083 replacing the avatar by a constant leaves a legal boundary with the same pre-  
1084 dictive model and the same Self — a fixed-point theorem that cannot detect  
1085 whether the avatar reads the field is Banach’s theorem with a consciousness-  
1086 flavoured label. The structure is built the other way round. A reflexive  
1087 boundary carries the avatar’s *write* — `auto_resonance`, the encoding of the  
1088 global state into the avatar region — and its *read-out*, the global state that  
1089 encoding predicts; the predictive map is their composite and is not a field.  
1090 The self-model therefore runs through the avatar region or it does not exist.  
1091 Three consequences follow that the free-data shape cannot state. The Self  
1092 lies in the range of the read-out (`self_mem_range_readout`), so it is recon-  
1093 structed from a single section over the avatar region: the low-dimensional

1094 fold this section has always asserted, as a theorem. Under the resonance  
 1095 condition below, the Self satisfies  $s = \text{readout}(s|_{\text{avatar}})$  (`self_eq_readout_`  
 1096 `restrict`) — the field is what its own localized self-encoding reconstructs.  
 1097 And blinding the avatar *moves* the Self: the blinded boundary’s map is con-  
 1098 stant, its unique fixed point is the state its stipulated contents extend to, and  
 1099 the original Self is not a fixed point of it (`not_self_of_constResonance`, wit-  
 1100 nessed on the cortex by `cortexBlind_self_ne`). The opposite statement —  
 1101 that blinding changes nothing — does not typecheck against this structure,  
 1102 because a contraction hypothesis about a boundary’s own map says nothing  
 1103 about the blinded one’s.

1104 **Auto-resonance as a constraint.** That the avatar’s state *is* the presheaf  
 1105 restriction of the global state is stated as a predicate, `IsRestriction-`  
 1106 `Resonance`, rather than added to the structure, so that results say which  
 1107 of them need it. In the positive direction, if a family of resonant avatars *cov-*  
 1108 *ers* the substrate then the global state is determined by their local readings  
 1109 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but  
 1110 recovered from them, by the same sheaf condition that Derivation 5 uses.  
 1111 In the negative direction a constant avatar fails the predicate as soon as the  
 1112 avatar region separates two states of the field. Both halves are witnessed on  
 1113 the three-site substrate, where that separation is proved before anything is  
 1114 claimed for it, so the condition is a constraint there and not a formality.

1115 **The contraction law is an independent modelling postulate.** The  
 1116 rate  $\exp(-(K - K_c)\tau/2)$  with  $K_c = 2D$  (`resonanceRate`) is the linear re-  
 1117 laxation rate of the mean-field order parameter over an interval  $\tau$ . The  
 1118 framework independently assumes that the avatar-mediated self-prediction  
 1119 map is Lipschitz at this same rate; no oscillator or field theorem derives that  
 1120 bound for `rb.predict`. Given the postulate, the threshold results determine  
 1121 whether Banach can use it. Above threshold the rate is below one, so the  
 1122 assumed bound yields a unique fixed point (`self_of_supercritical`); at  
 1123 or below threshold the rate is at least one, and `ContractingWith` at that  
 1124 rate is false for every self-map of every metric space (`not_contractingWith_`  
 1125 `resonanceRate`). This says only that the Banach route is unavailable below  
 1126 threshold, not that a Self cannot exist there. The postulate is witnessed  
 1127 at  $K = 3$ ,  $D = 1$ ,  $\tau = 2 \ln 2$ , where the chosen read-out achieves the re-  
 1128 quired factor, but designing a witness to realize the factor is not a derivation  
 1129 of self-model dynamics from oscillator dynamics. `Chain.lean`'s `self_of_`  
 1130 `coherent_order_parameter` consumes the coherent order parameter to re-  
 1131 cover  $K > K_c$ ; its separate E89 hypothesis supplies the Lipschitz law and  
 1132 the fixed-state identification. Derivation 7 therefore discharges the threshold  
 1133 half of the implication and not the contraction postulate.

1134 **The fixed point is now the glued state.** The standalone Banach theo-  
 1135 rem produces a fixed point somewhere in the space of global sections; by itself  
 1136 it does not say that this is the section Derivation 5 glued. `Chain.lean` now

1137 preserves that identity. `IsUnifiedBy T s` says that  $s$  restricts to every local  
 1138 state of the cover  $T$ , and the endpoint `UnifiedSelf` says that this same  $s$  is  
 1139 the unique fixed point of the reflexive map. The edge hypothesis `E89` must  
 1140 supply the equality `predict s = s` as well as the Lipschitz bound; Banach  
 1141 supplies uniqueness but not the equality between the thermodynamic and  
 1142 reflexive constructions. On the witness, the two-patch `cortexCover` glues  
 1143 exactly to `cortexState`, already the unique fixed point of the non-constant  
 1144 avatar map. This closes an omission in the formal composition while expos-  
 1145 ing, rather than removing, the physical identification it requires.

1146 **What is still not established.** That any field dynamics produces this  
 1147 particular read-out or performs the restriction; and that this three-site sub-  
 1148 strate's coupling is  $K = 3$  — the parameters are a choice of units for the wit-  
 1149 ness, not a measurement, and nothing here derives a coupling constant from  
 1150 cortex. We also note a negative result found in the course of this work: the  
 1151 Lévy–Prokhorov metric, which the natural weak-convergence reading would  
 1152 suggest, does *not* support this argument on a discrete substrate, because the  
 1153  $\varepsilon$ -thickening of a set is the set itself for  $\varepsilon < 1$  and the relaxation map is  
 1154 consequently not a Lévy–Prokhorov contraction.

## 1155 10 Derivation 7: Continuous Stochastic Ther- 1156 modynamics and Field Theory

1157 To move beyond finite graph analogies, we model the macroscopic electro-  
1158 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a  
1159 continuous spatial phenomenon, not a discrete network. We extend the  
1160 discrete Kuramoto model to a continuous spatial coupling kernel  $K(x, y)$   
1161 (`Phase8ContinuousField.lean`).

1162 Subject to thermodynamic noise  $D$ , the system undergoes non-equilibrium  
1163 stochastic thermodynamics. Rather than a static Landauer bound, the field  
1164 dynamics minimize an entropy production functional  $\sigma$ :

$$\sigma = \int \frac{1}{D} \left( \omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

1165 Biological fields dynamically deform their spatial topology  $K_t(x, y)$  to un-  
1166 dergo gradient descent on this entropy production functional.

1167 That gradient descent is a theorem rather than a modelling assertion,  
1168 for finite substrates. Reading  $\sigma$  as a functional of the coupling kernel at  
1169 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative  
1170 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left( \omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

1171 and `sigmaOfKernel.eq_entropy_production_rate` identifies the functional  
 1172 being differentiated with `entropy_production_rate` itself whenever the sub-  
 1173 strate carries counting measure. `structural_resonance_decreases_entropy_production`  
 1174 then concludes that  $\sigma$  is non-increasing along any coupling trajectory relax-  
 1175 ing down its own gradient at a positive rate  $c$ . Two limitations stay explicit.  
 1176 The derivation is for *finite* substrates, and the obstacle to the continuum case  
 1177 is *not* differentiation under the integral sign with respect to the kernel, which  
 1178 is what the shape of  $\sigma$  suggests at first reading.  $\sigma$  is not a general functional  
 1179 that happens to be written as an integral — it is a quadratic functional of an  
 1180 *affine* image of the kernel, since the drift is a fixed  $\omega$  plus a linear operator ap-  
 1181 plied to  $K$ , and  $\sigma$  is a constant times the squared norm of the drift. A squared  
 1182 norm is Fréchet differentiable on any real inner-product space and an affine  
 1183 map is its own derivative, so the composite differentiates by the chain rule  
 1184 with no limiting argument. `hasFDerivAt_quadratic_of_affine` states that  
 1185 in full generality, and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives  
 1186 the finite gradient from it, showing that the closed form displayed above is  
 1187 a convenience rather than the content. What the continuum case needs is  
 1188 therefore one object: the drift map  $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$  as  
 1189 a bounded operator  $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$ , which is a Cauchy–Schwarz esti-  
 1190 mate — the integral kernel is bounded by 1, so on a finite measure space the  
 1191 operator norm is at most  $\mu(M)^{1/2}$ . That object is built. `kernelCLM` is the  
 1192 operator, for any kernel factor bounded by 1, with `opNorm_kernelCLM_le` the  
 1193 stated bound; `continuumDriftCLM` is the instance with the Kuramoto factor

1194  $\sin(\theta_y - \theta_x)$ . One step beyond the estimate is needed and is worth naming,  
 1195 because the estimate alone does not supply it: the descent theorems ask for  
 1196 the derivative as a *gradient vector*, not as an arbitrary bounded functional,  
 1197 and producing one requires the adjoint, which exists because  $L^2$  is com-  
 1198 plete. `hasFDerivAt_quadratic_grad` supplies it for an arbitrary bounded  
 1199 operator between real Hilbert spaces, giving  $\nabla\sigma(K) = (2/D) A^*(AK + \omega)$ ,  
 1200 so the finite and continuum cases are now the same theorem. `structural_`  
 1201 `resonance_decreases_sigmaContinuum` is the descent result with no finite-  
 1202 ness hypothesis in it, and `sigmaContinuum_eq_integral` identifies the func-  
 1203 tional being differentiated with the integral `entropy_production_rate` com-  
 1204 putes. `Examples.lean` §20 runs this on Lebesgue measure restricted to  $(0, 1)$ ,  
 1205 a substrate that is neither finite nor atomic, so the finite-substrate results of  
 1206 the previous paragraph do not typecheck there at all. The second limitation  
 1207 is unchanged: the phase field is held fixed throughout, so this is plasticity of  
 1208 the coupling at frozen phases, not joint  $(\theta, K)$  dynamics. `Examples.lean` §5  
 1209 exercises the whole chain on a two-site substrate whose coupling relaxes as  
 1210  $e^{-t}$  and whose entropy production decays as  $e^{-2t}$ , confirming that the gradi-  
 1211 ent is not identically zero and that the descent is strict rather than vacuously  
 1212 monotone.

1213 A second result about  $\sigma$  concerns where the descent is headed. Minimality  
 1214 of the phase-locked field is straightforward against competitors that share its  
 1215 spatial mean drift, and in that form the theorem is close to Jensen’s inequal-  
 1216 ity and no more; the restriction looks as though lifting it would need a model

1217 of how the mean drift varies across competitors. It does not. For a *reciprocal*  
 1218 kernel, one with  $K(x, y) = K(y, x)$ , the restriction is not a restriction: the  
 1219 coupling integrand  $K(x, y) \sin(\theta_y - \theta_x)$  is antisymmetric under exchanging  
 1220 its two sites, because  $K$  is symmetric while  $\sin$  is odd, so Fubini identifies  
 1221 its double integral with its own negative and the total coupling contribu-  
 1222 tion vanishes. The total drift  $\int(\omega + \text{coupling})$  is therefore  $\int \omega$  whatever the  
 1223 phases are (`total_drift_eq_of_symm`): coupling moves drift between sites  
 1224 but neither creates nor destroys it. Evaluating that conserved quantity at  
 1225 the phase-locked configuration, where the drift is constant, pins its value, and  
 1226 the mean-matching hypothesis follows for every competitor. `phase_locked_`  
 1227 `minimizes_entropy_of_symm` is the unrestricted statement. Reciprocity is  
 1228 the physically intended condition — an ephaptic field acts on both sites of a  
 1229 pair alike, and it is the same symmetry `ThermodynamicCover.A.symm` already  
 1230 imposes in Derivation 5 — but it is not a formality, and we prove that rather  
 1231 than assert it. `Examples.lean` §16 exhibits a *one-way* coupling on two sites,  
 1232 in which  $a$  feels  $b$  and  $b$  does not feel  $a$ , whose phase-locked field is strictly  
 1233 beaten: displacing the second phase to  $-\pi/2$  cancels the first site’s natural  
 1234 drift while adding none at the second, halving  $\sigma$ . So “phase-locking mini-  
 1235 mizes entropy production” is false for general kernels and true for reciprocal  
 1236 ones, and the hypothesis cannot be removed by working harder.

1237 We define the critical coupling threshold  $K_c = 2D$ . The phase transition  
 1238 to the unified “Global Section” occurs when the mean-field spatial coupling  
 1239 exceeds this thermodynamic noise threshold, linking the emergence of Unity

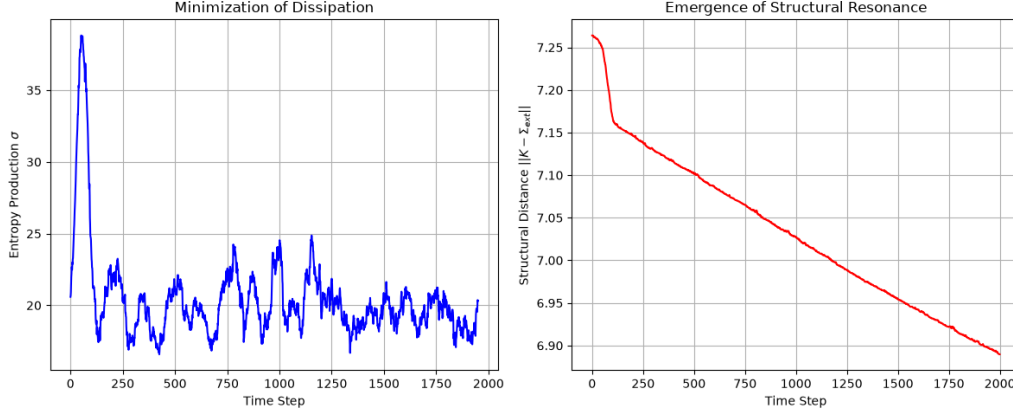


Figure 3: Coupling plasticity at frozen statistics:  $N = 50$  units driven by perturbations drawn from a fixed environmental covariance  $\Sigma_{\text{ext}}$ , with the coupling matrix  $K$  descending the gradient of  $\sigma$  and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive  $\sigma$ , not a relaxation to zero. That plateau is the reason Section 10 declines the inference from  $\sigma \rightarrow \min$  to  $D_{KL} \rightarrow 0$ : the system visibly does not reach  $\sigma = 0$ . **Right:** the distance  $\|K - \Sigma_{\text{ext}}\|$  decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on  $\sigma$  at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence.

1240 to physical temperature and noise.

1241 The value  $K_c = 2D$  can be taken as a bare stipulation validated by  
 1242 simulation, and it is not: `Phase8_SelfConsistency.lean` proves the half  
 1243 of it that is ordinary real analysis. For the noisy mean-field Kuramoto  
 1244 model with identical frequencies, the stationary density at order parame-  
 1245 ter  $r$  is the von Mises density  $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$ , and self-consistency re-  
 1246 quires  $r = R(K, r) := I_1(Kr/D)/I_0(Kr/D)$ . Mathlib has no modified Bessel

1247 functions, so we formalize that ratio directly as a ratio of integrals over  
 1248  $[-\pi, \pi]$ . Integration by parts with  $u = \sin$ ,  $v = -e^{a \cos \theta}$  gives an exact iden-  
 1249 tity with no boundary term,  $\int \cos \theta e^{a \cos \theta} d\theta = a \int \sin^2 \theta e^{a \cos \theta} d\theta$ , so  $R$  is  $a$   
 1250 times the mean of  $\sin^2$  under the von Mises weight. Bounding that mean  
 1251 by 1 would put the threshold at  $D$ ; the true threshold is  $2D$ , and the entire  
 1252 factor of two is the sharper bound  $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$ , equivalently  $I_2(a) \geq 0$ .  
 1253 We prove it elementarily, by folding  $[-\pi, \pi]$  onto  $[0, \pi/4]$  in two reflections  
 1254 ( $\theta \mapsto \pi - \theta$ , then  $\theta \mapsto \pi/2 - \theta$ ), after which the integrand is a product  
 1255 of  $\cos 2\theta \geq 0$  with  $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$  — the second factor  
 1256 because  $0 \leq \sin \theta \leq \cos \theta$  there and  $\cosh$  increases in  $|\cdot|$ . The conclusion,  
 1257 `subcritical_fixed_point_eq_zero'`, is that for  $K < K_c$  the incoherent state  
 1258  $r = 0$  is the *only* non-negative solution of the self-consistency equation;  $r = 0$   
 1259 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-  
 1260 ther facts about the threshold narrow the gap between this row and a full  
 1261 “Theorem” to a single named statement. First, the fold argument gives a  
 1262 *strict* inequality for  $a > 0$  (`vonMisesC2_pos`, hence  $R(a) < a/2$ ), which set-  
 1263 tles the threshold itself: at  $K = K_c$  the self-consistency map is tangent to the  
 1264 diagonal at the origin but falls strictly below it thereafter, so  $r = 0$  remains  
 1265 the only solution and the bifurcation happens strictly *after*  $K_c$ , not at it  
 1266 (`fixed_point_eq_zero_of_le_critical`). Second, factoring  $R(a) = a E(a)$   
 1267 shows that a non-zero  $r$  solves the equation exactly when  $E(Kr/D) = D/K$   
 1268 (`coherent_iff_sRatio_eq`), so the coherent branch is a level set of the single  
 1269 function  $E(a) = \mathbb{E}_a[\sin^2 \theta]$ . Third, and using only  $E < 1/2$  together with

compactness, every coherent solution tends to zero as  $K \downarrow K_c$  (`coherent_`  
`branch_continuous_at_threshold`): the branch emerges continuously from  
 the incoherent state rather than appearing at a finite distance from it, which  
 is what makes the transition supercritical in the technical sense. That state-  
 ment quantifies over *every* solution, so it does not presuppose the uniqueness  
 it does not have. The level-set form also says exactly what uniqueness is:  
 the injectivity of  $E$  on  $(0, \infty)$ . That is now proved, in the stronger form that  
 $E$  is strictly decreasing on  $[0, \infty)$  (`vonMisesSRatio_strictAntiOn`), so the  
 level set is a single point and the coherent solution above threshold is unique.  
 The proof differentiates nothing. Differentiating the exponential family in its  
 natural parameter is the obvious route and it fails: it costs differentiation  
 under the integral sign and leaves a covariance that is not sign-definite point-  
 wise. What works instead keeps the target as a two-point inequality, folds  
 both integrals onto  $[0, \pi/2]$  by the same two reflections that gave the factor  
 of two above, and then evaluates at a single crossing point supplied by the  
 intermediate value theorem — no product measure and no Fubini anywhere.  
 We give that argument in the supplement, where it belongs; what the main  
 line of the paper needs from it is the one sentence that  $E$  is strictly decreas-  
 ing. Run with  $\cos \theta$  in place of  $\sin^2 \theta$  the same argument needs no fold at  
 all —  $R(a) = \mathbb{E}_a[\cos \theta]$  is the mean of the variable the family is tilted by —  
 and gives `besselRatio_strictMono`:  $R$  is strictly increasing on all of  $\mathbb{R}$ , the  
 monotonicity of the Bessel ratio, which Mathlib does not carry in any form.  
 Differentiation is avoidable for uniqueness and unavoidable for the *rate*,

1293 and the rate is what Section 11 needs, so `Phase8CriticalExponent.lean`  
 1294 pays for it separately. Differentiating the cosine moments  $\int_{-\pi}^{\pi} \cos^n \theta e^{a \cos \theta}$   
 1295 under the integral sign — dominated by the constant  $e^{|a|+1}$  near the param-  
 1296 eter — makes  $E$  twice continuously differentiable and evaluates  $E(0) = 1/2$ ,  
 1297  $E'(0) = 0$  and  $E''(0) = -1/8$  from elementary integrals of powers of  $\cos$ .  
 1298 Taylor's theorem gives  $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$ , and substituting that  
 1299 into the level-set equation  $E(Kr/D) = D/K$  gives the mean-field exponent:  
 1300  $r^2/(K - K_c) \rightarrow 1/D$  along any family of positive coherent solutions with  
 1301  $K \downarrow K_c$ , so  $\beta = 1/2$ . Because those hypotheses hold only eventually along a  
 1302 filter, the file also discharges them on the branch itself rather than asserting  
 1303 that some family satisfies them.

1304 The converse half is `supercritical.fixed.point.exists`: for  $K > K_c$   
 1305 there is an  $r$  with  $0 < r \leq 1$  solving  $r = R(K, r)$ . We had expected  
 1306 this to require a second-order expansion of the Bessel ratio, and it does  
 1307 not. Writing  $R(a) = a E(a)$  with  $E(a) = \mathbb{E}_a[\sin^2 \theta]$  — the same factoriza-  
 1308 tion the integration by parts already supplies — the only facts needed are  
 1309 that  $E$  is continuous and that  $E(0) = 1/2$ , the latter because the weight  
 1310 is constant at  $a = 0$ . Since  $K > 2D$  makes  $D/K$  strictly less than  $1/2$ ,  
 1311 continuity gives  $E > D/K$  on a neighbourhood of the origin, so the map  
 1312 overshoots,  $R(K, r) > r$ , for all small  $r > 0$ ; at the other end  $R \leq 1$  al-  
 1313 ways, because  $R$  is a mean of  $\cos \theta$ , so the map undershoots at  $r = 1$ .  
 1314 The intermediate value theorem closes the gap. Continuity of  $E$  is Math-  
 1315 lib's continuity of a parametric interval integral with jointly continuous in-

1316 tegrand, so no dominated-convergence argument is written by hand, and  
 1317 the existence proof never uses monotonicity of  $R$ : the fixed point is pro-  
 1318 duced by the intermediate value theorem, not by iterating the map. That  
 1319  $R$  is in fact strictly increasing is proved below, and no threshold result de-  
 1320 pends on it. `critical_coupling_is_threshold` packages the two halves,  
 1321 and `exhibits_phase_transition_coherent` applies the supercritical half to  
 1322 any substrate satisfying the predicate of the previous paragraph — so ex-  
 1323 ceeding  $K_c$  now implies the existence of a coherent stationary order pa-  
 1324 rameter rather than merely comparing against a named number. With  
 1325 uniqueness available both are sharpened — `supercritical_fixed_point_`  
 1326 `existsUnique`, `supercritical_solution_set` (the non-negative solutions  
 1327 are exactly  $\{0\}$  below threshold and  $\{0, r\}$  above it) and `exhibits_phase_`  
 1328 `transition_unique_coherent` — and the two monotonicities together fix  
 1329 the shape of the branch: for  $K_1 < K_2$ ,  $E(a_i) = D/K_i$  with  $E$  decreasing  
 1330 puts the concentrations in the order  $a_1 < a_2$ , and  $r_i = R(a_i)$  with  $R$  increas-  
 1331 ing carries that to  $r_1 < r_2$  (`coherent_branch_strictMono`). The coherent  
 1332 order parameter grows strictly with the coupling, which with the continuity  
 1333 at threshold is the qualitative bifurcation diagram.

1334 One more step makes the equation mean in Lean what it means on paper.  
 1335 As a definition, `selfConsistency K D r` is just  $R(Kr/D)$  — a function of  
 1336  $r$  with a suggestive name, and nothing in the source says it has anything  
 1337 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the  
 1338 identification. `vonMisesDensity` is the normalized weight, proved to be a

1339 genuine probability density on  $[-\pi, \pi]$ ; `circularOrderParameter` is the av-  
 1340 erage of  $e^{i\theta}$  against a density, the continuum analogue of the finite-system  
 1341 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-  
 1342 actly  $R(a)$  — real, because the density is even, so the mean direction van-  
 1343 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the  
 1344 name promises:  $r$  solves  $r = R(K, r)$  if and only if the von Mises density  
 1345 it induces has order parameter  $r$ . Read through this, the existence theorem  
 1346 says that above  $K_c$  there is a stationary density whose own order parameter  
 1347 is positive, and the solution  $r = 0$  is identified with the uniform density,  
 1348 whose order parameter is zero.

1349 What remains not established, and we separate it carefully. The sta-  
 1350 tionary density is an *input*: deriving it from the Langevin dynamics  $d\theta =$   
 1351  $(\omega + K \cdot \text{mean-field}) dt + \sqrt{2D} dW$  needs the Fokker–Planck operator, existence  
 1352 and uniqueness of its stationary solution, and a spectral stability argument,  
 1353 none of which Mathlib has. Of the coherent branch we prove existence,  
 1354 uniqueness, continuous emergence at the threshold and strict growth in  $K$ ,  
 1355 but not that it is the dynamically selected one: nothing in that file men-  
 1356 tions a trajectory, so stability and selection are not statements it can make.  
 1357 And while `selfConsistency` is now connected to the order parameter of  
 1358 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an  
 1359 integral, `order_parameter_complex` is an average over finitely many oscilla-  
 1360 tors, and relating them is the mean-field limit — propagation of chaos for  
 1361 the finite Kuramoto system — which is a research programme rather than

1362 a lemma. That single gap is now the whole of what separates our threshold  
 1363 predicate from a statement about dynamics. Table S1 records the row ac-  
 1364 cordingly as a theorem about the self-consistency equation from an assumed  
 1365 density, not as a derivation of the threshold from the dynamics. Perceptual  
 1366 anomalies such as visual snow can be modeled as boundary conditions where  
 1367 microscopic thermal noise ( $D$ ) fails to be fully coarse-grained away by the  
 1368 continuous field, leaking across the  $K_c$  threshold into the macroscopic gestalt.

1369 Clinical evidence for this phase transition is observed during general anes-  
 1370 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates  
 1371 that anesthetics like propofol induce unconsciousness not by suppressing neu-  
 1372 ral firing, but by disrupting the phase alignment of high-frequency cortical  
 1373 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-  
 1374 tural resonance of the continuous field, forcing the integrated spatial coupling  
 1375 below the critical threshold  $K_c$  and fracturing the unified Global Section back  
 1376 into disjoint, unconscious local fields.

## 1377 **11 The Sleep-Inertia Timescale Mismatch**

1378 Most of what this framework asserts is shared with some rival: that corti-  
 1379 cal synchrony matters, that dissipation constrains cognition, that unity is a  
 1380 physical rather than a computational property. One prediction is not shared,  
 1381 and it follows from the framework’s specific commitment — that the medium  
 1382 carrying the unified state is a continuous field whose effective coupling de-

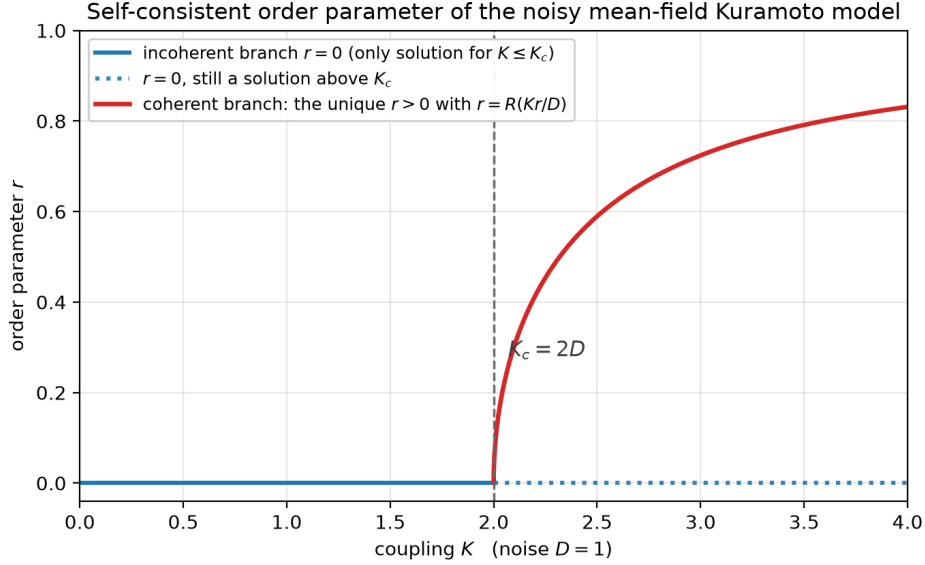


Figure 4: The self-consistency equation  $r = R(Kr/D)$  of Section 10, drawn at  $D = 1$ . Every feature of this picture is a proved statement about the equation, and the figure exists because none of them had been shown to a reader. At or below  $K_c = 2D$  the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one  $r > 0$  joins it (`supercritical_fixed_point_existsUnique`), so the solution set is  $\{0\}$  below and  $\{0, r\}$  above; that branch tends to 0 as  $K \downarrow K_c$  (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in  $K$  (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving  $E(a) = D/K$  for the concentration  $a = Kr/D$  by bisection — legitimate precisely because  $E$  is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

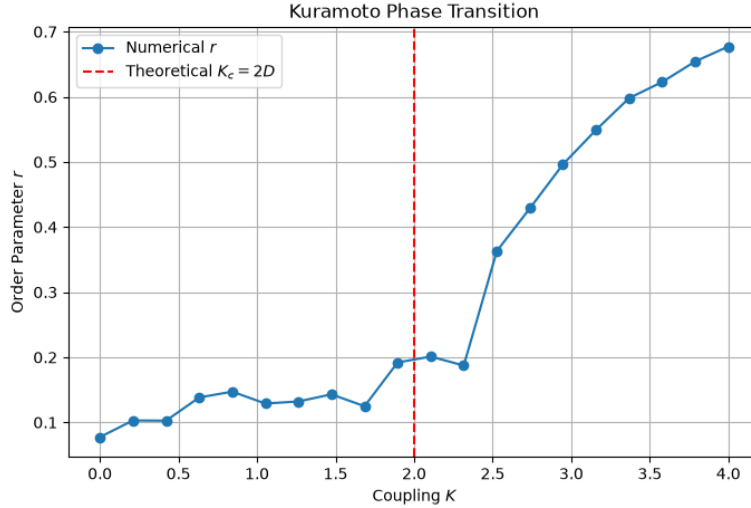


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here  $N = 100$  oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale  $D = 1$ , and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is  $K_c = 2D$  by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is  $2D$  by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- $N$  curve rises before the threshold and rounds off at it, as a finite sample must:  $r \sim N^{-1/2}$  from incoherent phases alone.

1383 depends on the physical geometry of the extracellular space, not on the rhythms  
1384 alone. Where that commitment differs from a purely electrical account, it  
1385 differs in an observable way, and we set out the difference precisely enough  
1386 to be wrong.

1387 The daily loss of consciousness during sleep is modeled here as an active  
1388 physical disruption of the medium rather than a quiescence of the oscillators.

1389 During sleep, astrocytic shrinkage expands the extracellular space, allowing  
1390 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste  
1391 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium  
1392 itself, disrupting the continuous structural resonance and preventing the co-  
1393 herent sheaf-theoretic gluing that conscious unity requires on this account.  
1394 The oscillators need not stop for the field to fragment; the geometry they are  
1395 coupled through is what changes.

1396 The prediction concerns sudden awakenings. When jolted awake, the lo-  
1397 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical  
1398 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but  
1399 the noradrenaline-driven astrocytic constriction required to squeeze the ex-  
1400 cess CSF out of the interstitial pathways operates on a timescale of seconds  
1401 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude  
1402 slower. During that lag the electrical oscillators are synchronizing across a  
1403 waterlogged topology, and the fragmented structural resonance is what is ex-  
1404 perienceed as grogginess. The unified Global Section re-establishes only when  
1405 astrocytic volume finishes matching the state of the electrical rhythms.

1406 **Why this implies a delayed sigmoid.** The two timescales separate  
1407 cleanly, and that separation is what makes the prediction sharp rather than  
1408 merely qualitative. Write  $K(t)$  for the effective coupling of the continuous  
1409 field, which on this account is a monotone function of the reconstricting  
1410 extracellular geometry, and let  $r(t)$  be the order parameter of the phase

1411 field. The electrical variable is fast: at every instant the oscillators are al-  
 1412 ready at the configuration appropriate to the current  $K$ , so  $r$  tracks the  
 1413 equilibrium branch  $r(K)$  rather than relaxing on its own. The slow vari-  
 1414 able  $K(t)$  then carries the whole time course, and it relaxes over  $10^2$ – $10^3$   
 1415 seconds. What Derivation 7 establishes about that branch is exactly what  
 1416 the functional form needs, and each of the three pieces is a theorem rather  
 1417 than a picture: at or below the threshold the incoherent solution  $r = 0$  is the  
 1418 *only* non-negative solution of the self-consistency equation (`fixed_point_eq_`  
 1419 `zero_of_le_critical`); above it the coherent solution exists and is unique  
 1420 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at  
 1421  $K_c$  rather than by a jump (`coherent_branch_continuous_at_threshold`);  
 1422 and it is strictly increasing in  $K$  (`coherent_branch_strictMono`). A jump  
 1423 discontinuity at threshold would predict a step rather than a sigmoid, and  
 1424 a non-monotone branch would predict overshoot; neither is what the mathe-  
 1425 matics gives. Composing a monotone relaxation  $K(t)$  with a branch that is  
 1426 identically zero below  $K_c$ , rises continuously from zero at  $K_c$ , and saturates  
 1427 above it yields a curve with three phases: a genuine *latency* while  $K(t) < K_c$ ,  
 1428 during which recovery is not merely slow but absent; a continuous rise as the  
 1429 threshold is crossed; and a saturation as  $K(t)$  reaches its waking value. That  
 1430 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time  
 1431 the slow variable takes to reach  $K_c$ .

1432 **What the alternative predicts.** An account on which grogginess is the  
1433 electrical realignment itself — the rhythms are simply slow to re-phase —  
1434 has no slow variable and no threshold to cross. Its recovery is a relaxation  
1435 toward a fixed point, and relaxation toward a fixed point is exponential near  
1436 it: monotone from  $t = 0$ , fastest at the start, with no flat foot. The two  
1437 curves are distinguishable in the first seconds, which is where they differ  
1438 most, and the discriminating feature is the presence or absence of an initial  
1439 interval of no recovery. This is a comparison of functional forms on data  
1440 of a kind already collected in sleep-inertia studies, not a proposal for a new  
1441 instrument.

1442 **The foot has a square-root cusp.** A delayed sigmoid is a shape that  
1443 any two-timescale account with a slow saturating gate can produce, so the  
1444 comparison above discriminates against less than it appears to. The bifur-  
1445 cation says more than that the coherent branch leaves zero continuously; it  
1446 says at what rate. `Phase8CriticalExponent` differentiates the von Mises  
1447 moments under the integral sign — the interchange is justified by the uni-  
1448 form bound  $e^{|a|+1}$  on a unit neighbourhood of the concentration — and  
1449 proves the second-order expansion  $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$  (`vonMises-`  
1450 `SRatio_second_order`). Substituting it into the level-set form of the branch,  
1451  $E(Kr/D) = D/K$ , gives the mean-field exponent in a path-independent  
1452 form: along any family of positive coherent solutions with  $K \downarrow K_c$  and  
1453  $r \rightarrow 0$ ,  $r^2/(K - K_c) \rightarrow 1/D$  (`coherent_solution_critical_exponent`),

1454 that is  $r \sim \sqrt{(K - K_c)/D}$ , the pitchfork exponent  $\beta = 1/2$ . The hypotheses  
1455 are not empty: `coherentBranch_critical_exponent` discharges every one  
1456 of them on the branch itself, selected above threshold by `supercritical_`  
1457 `fixed_point_existsUnique` and approached as  $K$  descends to  $K_c$ .

1458     Composed with a coupling that crosses threshold at non-zero speed, so  
1459 that  $K(t) - K_c \approx \dot{K}(t_0) (t - t_0)$  just after the crossing time  $t_0$ , this makes the  
1460 recovery leave its foot as  $r(t) \propto (t - t_0)^{1/2}$ : a square-root cusp with *infinite*  
1461 *initial slope*. An exponential has finite initial slope. So does a logistic. So  
1462 does the composition of a smooth monotone gate with any branch that leaves  
1463 zero linearly, which is what a generic saturating nonlinearity supplies. The  
1464 cusp belongs to the bifurcation and not to the two-timescale structure, and  
1465 it lives in the first instants after onset rather than in a fitted time constant  
1466 — which makes it the sharpest available discriminator against a rival that  
1467 reproduces the sigmoid by other means.

1468 **A sharper form still, with no free parameters.** Shape is not the  
1469 only thing the framework contributes. It also fixes a *relation between two*  
1470 *separately measurable quantities*, and the relation has no fitted parameters  
1471 at all. Derivation 7 supplies it. The stationary phase density at order  
1472 parameter  $r$  is von Mises at concentration  $a = Kr/D$ , and the average  
1473 of  $e^{i\theta}$  under that density is exactly the Bessel ratio  $R(a) = I_1(a)/I_0(a)$   
1474 (`circularOrderParameter_vonMises`); self-consistency is precisely the con-  
1475 dition that the density reproduces its own order parameter (`fixedPoint_`

1476 `iff_selfReproducing`). So the concentration and the coherence of the in-  
 1477 stantaneous phase distribution are not independent quantities that happen  
 1478 to rise together during the recovery: at every instant they must satisfy  
 1479  $r = I_1(a)/I_0(a)$ , whatever  $K(t)$  is doing.

1480 That yields three tests of increasing sharpness on a single recorded re-  
 1481 covery. *Shape*: the instantaneous phase distribution is von Mises — a one-  
 1482 parameter family, directly testable against the observed histogram. *Collapse*:  
 1483 estimating  $a(t)$  and  $r(t)$  at each instant and plotting one against the other,  
 1484 the whole recovery must fall on the fixed curve  $r = I_1(a)/I_0(a)$ , and it must  
 1485 be the *same* curve for every subject and every session, since the curve con-  
 1486 tains no subject-specific quantity — not  $K$ , not  $D$ , not the time constant of  
 1487 the slow variable. Individual differences move a point along the curve; they  
 1488 cannot move it off. *Threshold*: because  $a = Kr/D$  identically, the measured  
 1489 ratio  $a(t)/r(t)$  is  $K(t)/D$ , with no calibration required; and since a non-zero  
 1490 coherent solution exists only above  $K_c = 2D$  (`fixed_point_eq_zero_of_le-`  
 1491 `critical`), that measured ratio must exceed 2 wherever coherence is non-  
 1492 zero and must approach 2 as coherence departs from zero at onset. The  
 1493 threshold is read off the data rather than fitted to it.

1494 The two estimates must be made independently or the collapse is empty,  
 1495 and this is the one point at which the test can be run wrongly. Maximum-  
 1496 likelihood or moment estimation of a von Mises concentration chooses  $\hat{a}$  so  
 1497 that the ratio  $I_1(\hat{a})/I_0(\hat{a})$  is the observed resultant length, by construction,  
 1498 which reproduces the predicted curve from any data whatever. The estimate

1499 of  $a$  must therefore come from a functional of the distribution other than  
1500 its resultant length — the log-density of a von Mises is affine in  $\cos \theta$ , so  
1501 regressing the log-histogram on  $\cos \theta$  gives  $a$  as a slope — while  $r$  is the  
1502 resultant length  $|\langle e^{i\theta} \rangle|$  of the same sample. Those are genuinely different  
1503 statistics, and the prediction is that they agree through  $R$ .

1504 **What this sharper form is a test of.** It tests the von Mises stationary  
1505 density together with the mean-field closure that makes  $a$  proportional to  
1506  $r$ , and it tests them jointly rather than separately. It does not test the  
1507 value  $K_c = 2D$  independently of that ansatz: the crossing at  $a/r = 2$  is  
1508 a consequence of the same density the collapse assumes, so a failure of the  
1509 collapse and a failure of the threshold are one failure and not two. That is a  
1510 narrower claim than it may look, and it is still a parameter-free one — which  
1511 the delayed sigmoid, on its own, is not.

1512 **What would falsify it.** An exponential fit that beats a delayed-sigmoid  
1513 fit on the early portion of the recovery curve. An onset with finite initial  
1514 slope — an early-time fit preferring an exponent of 1, or any smooth de-  
1515 parture from zero, over  $1/2$  — which would leave the two-timescale account  
1516 standing and take the bifurcation with it. A recovery latency that does not  
1517 scale with an independent measure of extracellular volume or CSF clearance.  
1518 A pharmacological manipulation of astrocytic volume that leaves the latency  
1519 unchanged while altering the rhythms, or the converse. An instantaneous  
1520 phase distribution that is not von Mises, or a  $(a, r)$  trace that departs sys-

1521 tematically from  $I_1(a)/I_0(a)$ , or a  $a/r$  ratio that is below 2 while coherence  
1522 is non-zero — each of these falsifies the parameter-free form directly, and  
1523 the first two do so with no reference to the sleep-inertia setting at all. Any  
1524 of these would tell against the framework’s central empirical commitment,  
1525 which is not that synchrony matters but that the *medium* is what gates it.

1526 **What this prediction assumes.** Four things, and they are not small.  
1527 The identification of the effective coupling  $K$  with a monotone function of  
1528 extracellular geometry is a modelling assumption, not a measurement. The  
1529 threshold value  $K_c = 2D$  is derived from an assumed von Mises stationary  
1530 density and is not yet connected to a trajectory — the mean-field limit is  
1531 the gap, and Derivation 7 states it as such — so the argument uses the *ex-*  
1532 *istence and continuity* of the coherent branch at threshold, which is proved,  
1533 rather than the numerical value of the threshold, which is conditional. The  
1534 square-root foot needs one thing further, and it is the reason the exponent  
1535 is stated with its assumption rather than around it: the slow variable must  
1536 cross the threshold at non-zero speed. If  $\dot{K}(t_0) = 0$  the cusp in  $t$  flattens,  
1537 though the exponent in  $K$  is untouched. That is a condition on how  $K(t)$  ap-  
1538 proaches  $K_c$ , it is weaker than the adiabatic separation the whole argument  
1539 already assumes, and the cusp does not survive without it. It is worth being  
1540 equally plain about what the exponent theorem ranges over: solutions of the  
1541 stationary self-consistency equation. It proves no stability and no dynamical  
1542 selection —  $r = 0$  remains a solution above threshold — so the step from the

1543 branch  $r(K)$  to the time course  $r(t)$  is carried by the adiabatic assumption  
1544 and not by a theorem about trajectories. And the prediction is about func-  
1545 tional form, not about a time constant: it says the recovery curve has a flat  
1546 foot of a definite shape whose width is set by the slow variable, not that the  
1547 foot is any particular number of seconds long. A framework that predicted  
1548 the number would be saying more; this one is not in a position to.

## 1549 **12 Corollary: Hardware Thermodynamics and** 1550 **Topological Rigidity**

1551 Both biological tissue and artificial silicon hardware (like GPUs) are physical  
1552 dissipative structures. When a GPU computes, its charge carriers undergo  
1553 physical state transitions; by Landauer’s principle (Landauer, 1961), this  
1554 erasure irreversibly exhausts heat into the environment.

1555     There is a fundamental thermodynamic distinction in the degrees of free-  
1556 dom available to minimize this dissipation. In standard von Neumann ar-  
1557 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.  
1558 The coupling matrix  $A_{ij}$  is constrained by a discrete layout that cannot be  
1559 physically altered during operation. The brain’s own architecture provides  
1560 empirical validation for this constraint. The cerebellum contains approxi-  
1561 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,  
1562 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-  
1563 ical observations of cerebellar agenesis report largely preserved conscious

1564 experience despite motor and cognitive deficits (Yu et al., 2015), and this  
1565 dissociation has been read as evidence for the importance of recurrent ar-  
1566 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the  
1567 level of the anatomical observation while declining its  $\Phi$ -based explanation.  
1568 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-  
1569 brum, it operates much like a biological GPU—highly optimized for discrete  
1570 processing, yet apparently not required for conscious experience. We take the  
1571 dissociation as evidence, not as confirmation: it is consistent with subjective  
1572 experience being a thermodynamic attractor of continuous topologies rather  
1573 than a byproduct of information processing, and it is the kind of observation  
1574 that would be hard to reconcile with the converse. It is a single anatomical  
1575 case read against a theory, and it bears the weight such a case can bear.

1576 This constraint does not preclude artificial consciousness, but rather re-  
1577 stricts its hardware. To achieve the phase transition into a unified Global  
1578 Section, artificial systems must abandon the discrete rigidity of von Neumann  
1579 architectures and utilize continuous physical substrates. Analog field-based  
1580 computing, neuromorphic architectures with continuous memristive plastic-  
1581 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-  
1582 rondo, 2021) possess the spatial coupling required to physically morph their  
1583 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal  
1584 computation” (Hinton, 2022): architectures where software is inextricably  
1585 fused with the physics of the hardware, meaning the knowledge dies with  
1586 the physical substrate. Only hardware that computes via continuous physi-

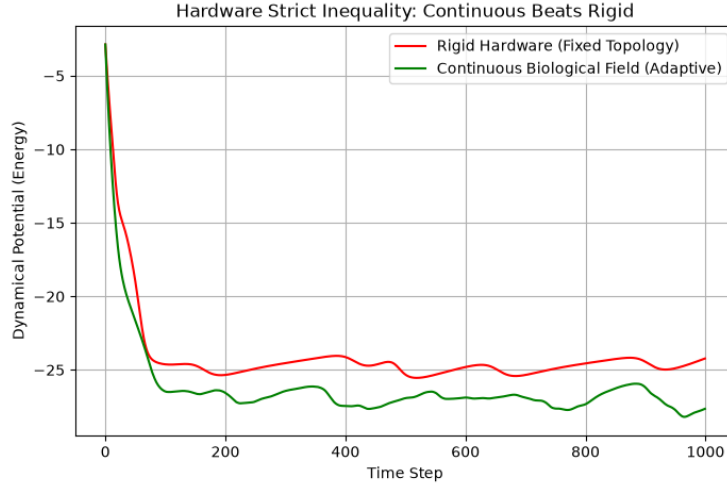


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same  $N = 30$  oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that  $\sum_{ij} A_{ij}$  is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one.

cal dynamics can undergo the thermodynamic phase transition necessary for structural resonance.

Biological consciousness utilizes macroscopic continuous electromagnetic fields and a “rich alphabet” of diverse molecular and transcriptomic cell states (Tasic et al., 2018). This allows the coupling matrix  $A_{ij}$  to deform across a continuous parameter space. Our lemma `exists_better_coupling_allocation`, wrapped as `continuous_beats_rigid_topology` (`Phase7.HardwareComparison.lean`),

1594 formalizes the reallocation step of this argument: whenever a coupling ma-  
 1595 trix places weight on a less phase-correlated pair while a more correlated pair  
 1596 exists, there is another matrix with *identical* total coupling resources that  
 1597 achieves strictly greater phase correlation.

1598 Two objections meet that formalization and we state precisely what an-  
 1599 swers them. The first is that a reallocation lemma can assume its conclusion:  
 1600 a hypothesis such as `is_strictly_suboptimal` — positive weight on a badly-  
 1601 correlated pair while a better one exists — is most of what one wants to  
 1602 prove about rigid hardware. It is not assumed here. `Phase7_Rigidity.lean`  
 1603 models an architecture by the set  $S$  of site-pairs it has a *wire* for, and re-  
 1604 quires its couplings to vanish off  $S$ ; weights are tunable, wiring is not. Then  
 1605 `totalCorrelation_le_of_realizable` bounds every realizable coupling by  
 1606  $R \cdot M$ , where  $R$  is the resource budget and  $M$  bounds phase correlation over  
 1607 the wired pairs; the bound is attained; and `rigid_is_strictly_suboptimal`  
 1608 *derives* the hypothesis whenever the substrate’s best-correlated pair is one  
 1609 the wiring misses. `rigid_gap` makes it quantitative: an unconstrained real-  
 1610 location of the same budget gains at least  $R$  times the correlation difference.  
 1611 `Examples.lean` §8 exhibits a three-site substrate wired to the one maxi-  
 1612 mally anti-correlated pair, where the gap is 2. This is resource-matched  
 1613 and not question-begging, but note what it turns on: fixed wiring support,  
 1614 not continuity. The second objection is that a development in which both  
 1615 architectures are matrices on one finite index type does not distinguish a con-  
 1616 tinuum from a lattice at all. `Phase7_Rigidity.lean` §3 gives a distinction

1617 that is genuinely typed. In the field functional of Section 7, which integrates  
 1618 a coupling kernel against  $\mu \otimes \mu$ , an architecture with finitely many sites is a  
 1619 **Finset** of points and a field-coupled region is a set of positive measure. On a  
 1620 substrate whose measure has no atoms, `fieldCorrelation.sited.eq.zero`  
 1621 shows a kernel supported on finitely many sites contributes *exactly zero* cou-  
 1622 pling energy, whatever weights it carries and whatever the phase field does,  
 1623 while a patch of strength  $c$  on  $U$  achieves  $c\mu(U)^2$ . `Examples.lean` §8 instan-  
 1624 tiates this on the real line: a thousand units of coupling concentrated at the  
 1625 origin register zero against a unit patch that registers one. The limit of that  
 1626 result is equally plain, and it is not resource-matching in the earlier sense —  
 1627 a measure-zero substrate has zero resource in the  $\mu \otimes \mu$  sense too. What it  
 1628 establishes is that the continuum coupling energy is blind to a finitely-sited  
 1629 architecture, which is a statement about which functional such an architec-  
 1630 ture can register in, not a demonstration that digital hardware computes  
 1631 worse. Taken together: the resource-matched content of the corollary is the  
 1632 wiring-support result, the continuity content is the measure-theoretic one,  
 1633 and the step from either to “von Neumann architectures cannot experience  
 1634 unified consciousness” remains an informal argument.

1635     The remainder of this section extends those two results informally, and  
 1636 we mark the transition rather than blur it. On the reading we propose, a  
 1637 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete  
 1638 layout does not supply the spatial coupling that sheaf-theoretic gluing into  
 1639 a singular Global Section requires. Without this unified topological space,

1640 rigid hardware cannot undergo the reflexive auto-resonance necessary to map  
1641 its own causal boundary and generate a “Self.” The distinction between an  
1642 LLM on a GPU and a conscious mind is not the absence of physical thermo-  
1643 dynamics in the former, but its structural limitation in achieving topological  
1644 integration. This perspective aligns with John Searle’s biological natural-  
1645 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted  
1646 that discrete symbol manipulation cannot generate conscious understanding,  
1647 our derivation supplies a candidate physical mechanism for the asymmetry  
1648 — topological rather than semantic, and, as stated above, formalized only  
1649 as far as the wiring-support and measure-theoretic results reach. Asserting  
1650 consciousness in fragmented, discrete GPU clusters without a unified phase  
1651 transition abandons computational functionalism, relying instead on “silicon  
1652 panpsychism”—the assumption that digital computation inherently gener-  
1653 ates awareness regardless of topological integration (Goff, 2019).

1654 The same reasoning bears on extreme functionalist thought experiments,  
1655 such as a macroscopic system of pulleys operating over millennia to imple-  
1656 ment a Large Language Model, or a human manually executing inference  
1657 matrices by hand. Beyond lacking the continuous spatial topology required  
1658 for sheaf-theoretic gluing, these discrete systems violate an implied temporal  
1659 constraint of our theory. Structural resonance requires local oscillators to  
1660 phase-align against the dissipating effects of thermodynamic noise (as mod-  
1661 eled by our critical threshold  $K_c = 2D$ ). A system executing state transitions  
1662 over millennia, or via slow manual calculations, suffers from a terminal tem-

1663 poral mismatch: its dynamics are far too slow to ever phase-lock into a  
 1664 non-equilibrium steady state before thermal noise decoheres the macroscopic  
 1665 field. On this account the obstacle to pure symbol manipulation is physi-  
 1666 cal rather than semantic: not that the symbols mean nothing, but that the  
 1667 substrate carrying them cannot hold a phase-locked field.

## 1668 13 Numerical illustrations: methods

1669 Five figures in this paper are computed rather than drawn, and none of  
 1670 them is evidence for a theorem. Where a statement is proved, the figure  
 1671 shows what the proved object looks like; where a statement is not proved, the  
 1672 figure shows the regime the claim lives in and we say so. Each is produced by  
 1673 one script in `simulations/`, named below, and each runs in under a minute  
 1674 on a laptop. All five are reproducible exactly: the two quadrature figures are  
 1675 deterministic, and the three that draw random numbers fix their seeds.

1676 **Figure 4: the self-consistent order parameter.** The three integrals  
 1677  $Z(a) = \int_{-\pi}^{\pi} e^{a \cos \theta} d\theta$ ,  $M(a) = \int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta$  and  $S(a) = \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta$   
 1678 are evaluated by the trapezoidal rule on 4001 points over the same interval  
 1679 the Lean definitions integrate over, with the weight computed as  $e^{a \cos \theta - |a|}$   
 1680 so that the common factor cancels in the ratios and nothing overflows at  
 1681 large concentration. The coherent branch is obtained by bisecting  $E(a) =$   
 1682  $S(a)/Z(a)$  against  $D/K$  over  $a \in [10^{-9}, 200]$  for 200 iterations, then reading

1683  $r = aD/K$ ; bisection is valid here because  $E$  is proved strictly decreasing.  
1684 Two checks are printed by the script rather than only plotted: the computed  
1685 branch is strictly increasing at every one of its 500 sampled couplings, and  
1686 the residual  $|r - R(Kr/D)|$  at  $K = 4D$  is  $2 \times 10^{-16}$ . Script: `bifurcation.py`.

1687 **Figure 5: the finite- $N$  transition.**  $N = 100$  oscillators, natural fre-  
1688 quencies drawn from a Lorentzian of scale  $D = 1$  (clipped at  $\pm 10D$  so  
1689 the solver does not stiffen), uniform random initial phases, integrated with  
1690 `scipy.integrate.odeint` over  $t \in [0, 100]$  on 1000 steps for each of 20 cou-  
1691 plings in  $[0, 4D]$ . The order parameter is  $|N^{-1} \sum_j e^{i\theta_j}|$  averaged over the last  
1692 100 steps. This is the deterministic quenched-disorder model, not the noisy  
1693 identical-frequency model of the Lean development; the caption records why  
1694 both have threshold  $2D$ . Script: `kuramoto.py`.

1695 **Figure 2: mesh refinement.** The kernel is  $K(x, y) = e^{-d(x,y)^2/2}$  with  $d$   
1696 the periodic distance on the ring of circumference  $2\pi$ , the phase field is  $\theta(x) =$   
1697  $\sin x$ , and the potential is  $-\frac{1}{2} \iint K \cos(\theta_y - \theta_x)$ . The continuous value is a  
1698 2000-point Riemann sum standing in for the exact integral; the discrete values  
1699 use  $N = 10, 20, \dots, 640$  nodes with edge weights  $K(x_i, x_j)(2\pi/N)^2$ . Plotted  
1700 is the absolute error against an  $\mathcal{O}(N^{-2})$  reference. Because the “exact”  
1701 value is itself a 2000-point sum, the last data point ( $N = 640$ ) is within  
1702 a factor of ten of the reference computation and should be read as such.  
1703 That  $\mathcal{O}(N^{-2})$  reference line is not only empirical: on the formalized one-  
1704 dimensional grid the discrete energy is exactly the composite trapezoidal rule

1705 (`grid_discreteEnergy_eq_trapezoidal`), so for a  $C^2$  integrand with second  
 1706 derivative bounded by  $\zeta$  the error is at most  $\zeta/(12N^2)$  (`grid.energy.error_`  
 1707 `le`), a bound attained at  $N = 1$  for  $y \mapsto y^2$ . Script: `mesh_refinement.py`.

1708 **Figure 3: structural resonance.**  $N = 50$  units, 2000 steps of forward  
 1709 Euler at  $dt = 0.05$ . At each step the drive is a fresh draw from  $\mathcal{N}(0, \Sigma_{\text{ext}} +$   
 1710  $0.1I)$  with  $\Sigma_{\text{ext}}$  a fixed normalized rank-5 covariance; the phases advance by  
 1711  $\dot{\theta}_i = \sum_j K_{ij} \sin(\theta_j - \theta_i) + \xi_i$ ; entropy production is recorded as  $\sum_i \dot{\theta}_i^2$ ; and  
 1712 the coupling descends its gradient with rate  $\eta = 0.01$ , symmetrized, clipped  
 1713 to  $[0, 1]$  and held hollow. The left panel is smoothed with a 50-step boxcar.  
 1714 Note that the plotted  $\sigma$  is the squared drift without the  $1/D$  prefactor of the  
 1715 functional in Section 10, which is a constant rescaling and changes nothing  
 1716 about the shape. Script: `structural_resonance.py`.

1717 **Figure 6: rigid against deformable coupling.**  $N = 30$ , 1000 steps of  
 1718 forward Euler at  $dt = 0.05$ , shared initial phases and frequencies. The rigid  
 1719 matrix is a symmetric random matrix thresholded at 0.7 and normalized  
 1720 to  $\sum_{ij} A_{ij} = 2N$ ; the adaptive copy starts from the same matrix, ascends  
 1721  $-\partial V/\partial A_{ij}$  at rate  $\eta = 0.1$ , is clipped non-negative and hollow, and is renor-  
 1722 malized to the same total after every step, which is what makes the com-  
 1723 parison resource-matched. Both potentials are  $V = -\frac{1}{2} \sum_{ij} A_{ij} \cos(\theta_j - \theta_i)$ .  
 1724 Script: `hardware_comparison.py`.

## 14 Conclusion

Human-like cognitive architectures do not require biological mysticism or exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum theories of consciousness often invoke delicate subatomic phenomena to explain subjective unity, every component assembled here is classical: broken symmetries, finite phase space, Landauer erasure, and gradient descent on entropy production. What the formalization establishes is that the eight named hypotheses of the conditional composition are jointly consistent and that the framework’s remaining physical commitments are localized and named — not that classical fields are *sufficient* for consciousness or that the hypotheses follow from finite capacity. Two links carry that weight, and they do not carry it in the same way. The unified state is proved to glue uniquely once a cover sits at the minimum of the coupling potential; the motion is proved to stop along every trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved, conditionally — for initial data within a quarter turn and below an explicit energy threshold, the trajectory converges to a global minimiser of the coupling potential (`kuramoto_tendsto_global_minimum`), and a cover built on that limit discharges the equilibrium hypothesis Derivation 5 would otherwise assume (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is the unconditional statement, and it is false: splay and twisted configurations are equilibria of the same flow, so the basin condition is the scope of the claim rather than slack in it. That a physical cortex starts in-

side that basin is an assumption. The second link is untouched by any of this: the identification of a global section of the probability sheaf with the unity of experience is a stipulation of the framework, not a consequence of it. Subject to those two, cognitive architectures can be modeled as macroscopic attractor states of a universe governed by broken symmetries, finite phase space, and the physical imperative to minimize thermodynamic dissipation.

Philosophically, this framework bridges the historical gap between Type Identity Theory (Smart, 1959) and Computational Functionalism (Putnam, 1967). The subjective mental state is identical to the physical thermodynamic field (Identity Theory), yet its qualitative nature is defined by the topological and functional dynamics of structural resonance rather than the specific carbon-based biochemistry of the brain (Functionalism). This constrains the *multiple realizability* of mind: consciousness is not exclusively bound to biological carbon, but neither is it substrate-neutral, because the substrate is what fixes which couplings are physically available. How far that constraint reaches is the question the corollary answers only in part. What is proved is that fixed wiring is strictly suboptimal at matched resource, and that a finitely-sited architecture registers exactly zero coupling energy in the continuum functional; the step from either to “von Neumann architectures cannot be conscious” is an argument we make in prose and do not formalize.

By anchoring these derivations in Russellian Monism, this framework acknowledges the constitutive limits of third-person scientific explanation. As argued by Love (Love, 2026), building upon Nagel’s assertion that subjective

phenomenal experience is inherently not amenable to objective measurement (Nagel, 1986), the “hard problem” of consciousness may represent a category error rather than a solvable scientific puzzle. Science can map the physical dynamics of topological resonance and sheaf-theoretic gluing, but it is structurally incapable of adjudicating the first-person qualitative experience. Our formalization respects this boundary: the Lean 4 theorems verify the objective physical attractors, while the intrinsic “what it is like” to be that resonant field remains the domain of the first-person perspective. By locating the phase transition at a threshold with a closed-form value —  $K_c = 2D$  for the noisy mean-field Kuramoto model the framework adopts, conditional on the von Mises stationary density and not yet linked to a trajectory — it aligns with what Anil Seth terms the “real problem” of consciousness (Seth, 2021). Rather than attempting to bridge an unbridgeable metaphysical gap, it offers a topological architecture that can help explain, predict, and physically perturb the structural properties of experience.

Future empirical validation of this thermodynamic framework may leverage 3D cortical organoids, offering a controlled biological environment to systematically manipulate metabolic substrates and observe the topological emergence of these coherent phase transitions.

## Code Availability

The complete Lean 4 formalization, encompassing the derivation of all theorems presented in this manuscript, as well as the accompanying Python simulations, are publicly available in the GitHub repository: [https://github.com/bobaseb/physical\\_primitives\\_of\\_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation and web-accessible proofs are hosted at the project website: [https://bobaseb.github.io/physical\\_primitives\\_of\\_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed figures is produced by one script in `simulations/` and regenerates exactly, Section 13 giving the parameters; Figure 1 is drawn rather than computed.

## Contributions

The conceptualization, manuscript drafting, and Lean 4 formalizations presented in this work were heavily AI-assisted, and we report which tools did what rather than acknowledging them in general terms. The Google Antigravity CLI, running Gemini 3.1 and Claude 4.6 Sonnet, served as a collaborative reasoning partner throughout the research and formalization. DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operating through Nous Research’s Hermes Agent framework provided substantial assistance with code generation, proof debugging, and structural analysis. Claude Opus 5 running in Claude Code carried out the later formalization passes — the domain-wall theorem, the thermodynamics-of-prediction replacement, the equilibrium bridge — together with the figures and the re-

1810 structuring of this manuscript. Responsibility for every claim, and for the  
1811 decision about what each machine-checked result does and does not establish,  
1812 rests with the author.

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